

TEST SPEC V1 — Section 1

Document Overview & Testing Scope

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Document Type:

Verification & Validation Specification

Version:

V1

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APPROVED

Classification:

Quality Assurance Artifact

Parent Documents:

- PRD V1

- ADD V1

- DDD V1

- SDD V1

- IAD V1
- TDD V1
- TRM V1

1.1 Purpose

This document defines the testing and validation strategy for TODP.

The purpose of TEST SPEC V1 is to:

- Verify requirement implementation
- Validate business functionality
- Validate architectural compliance
- Validate security controls
- Validate operational readiness
- Validate production readiness

The document serves as the authoritative source for all verification activities.

1.2 Testing Objectives

TEST-OBJ-01

Verify functional correctness.

TEST-OBJ-02

Verify requirement compliance.

TEST-OBJ-03

Verify architectural compliance.

TEST-OBJ-04

Verify security controls.

TEST-OBJ-05

Verify operational readiness.

TEST-OBJ-06

Verify production readiness.

1.3 Scope

This document covers testing activities for:

Authentication

Authorization

Opportunity Management

Authority Management

Evidence Management

Retrieval

Ranking

Recommendation

Governance

Audit

Security

Operations

Deployment

1.4 Testing Coverage Model

Requirements

↓

Design

↓

Implementation

↓

Testing



Verification



Approval

All approved requirements must be validated.

1.5 Testing Principles

TP-01

Testing must be repeatable.

TP-02

Testing must be measurable.

TP-03

Testing must be auditable.

TP-04

Testing must be reproducible.

TP-05

Testing must be traceable.

TP-06

Testing must support governance.

1.6 Verification Philosophy

Every requirement must have:

Test Coverage

↓

Execution Evidence

↓

Verification Result



Approval Record

No requirement may remain unverified.

1.7 Testing Domains

Functional Testing

Integration Testing

API Testing

Database Testing

Security Testing

Performance Testing

Operational Testing

Acceptance Testing

1.8 Test Levels

Level 1

Unit Testing

Level 2

Integration Testing

Level 3

System Testing

Level 4

Acceptance Testing

Level 5

Operational Validation

1.9 Verification Categories

Functional Verification

Technical Verification

Security Verification

Operational Verification

Governance Verification

Compliance Verification

1.10 In Scope Components

Frontend

Backend

Database

APIs

Services

Repositories

Caching Layer

Infrastructure

Deployment Pipelines

Monitoring Stack

1.11 Out of Scope Components

Third-Party Vendor Internal Systems

Internet Infrastructure

Cloud Provider Internal Services

External User Devices

1.12 Requirement Sources

Requirements originate from:

PRD V1

ADD V1

DDD V1

SDD V1

IAD V1

TDD V1

TRM V1

1.13 Requirement Verification Sources

Verification evidence includes:

Test Cases

Execution Reports

Coverage Reports

Security Reports

Performance Reports

Audit Records

1.14 Testing Lifecycle

Test Planning

↓

Test Design



Test Development



Test Execution



Defect Resolution



Retesting



Approval

1.15 Test Artifact Types

Test Plans

Test Cases

Test Suites

Test Data

Execution Reports

Coverage Reports

Defect Reports

Verification Reports

1.16 Traceability Requirements

Every test must trace to:

Requirement



Design Artifact



Implementation Component



Verification Result

1.17 Quality Attributes

The testing program validates:

Correctness

Reliability

Availability

Performance

Security

Maintainability

Auditability

1.18 Success Criteria

Testing is considered successful when:

Requirements Verified



Defects Resolved



Coverage Targets Achieved



Governance Approved

1.19 Stakeholders

Product Owner

Architect

Technical Lead

Developers

QA Engineers

Operations Team

Governance Team

Auditors

1.20 Test Governance

Testing activities must support:

Traceability

Auditability

Compliance

Reproducibility

Approval Workflows

1.21 Compliance Objectives

Verify compliance with:

Business Requirements

Architecture Requirements

Technical Requirements

Security Requirements

Operational Requirements

Governance Requirements

1.22 Testing Deliverables

Deliverables Include:

Test Cases

Test Suites

Execution Reports

Coverage Reports

Defect Reports

Compliance Reports

Verification Reports

1.23 Coverage Targets

Requirement Coverage:

100%

Critical Requirement Coverage:

100%

Security Requirement Coverage:

100%

Governance Requirement Coverage:

100%

1.24 Auditability Requirements

All testing activities must support:

Audit Reviews

Coverage Reviews

Compliance Reviews

Governance Reviews

1.25 Reproducibility Requirements

Testing must support:

Test Replay

Execution Replay

Result Replay

Regression Replay

Verification Replay

1.26 Governance Rules

TEST-GOV-01

Every requirement requires test coverage.

TEST-GOV-02

Every critical feature requires verification.

TEST-GOV-03

Testing evidence must be retained.

TEST-GOV-04

Coverage metrics must be measurable.

TEST-GOV-05

Testing activities must be auditable.

TEST-GOV-06

Testing activities must be reproducible.

1.27 Architectural Constraints

AC-TEST-01

Untested requirements are prohibited.

AC-TEST-02

Orphan test cases are prohibited.

AC-TEST-03

Critical functionality cannot bypass testing.

AC-TEST-04

Verification evidence cannot be deleted.

AC-TEST-05

Coverage gaps require remediation.

AC-TEST-06

Governance controls cannot be bypassed.

1.28 Section Summary

This section establishes the purpose, scope, objectives, governance model, testing philosophy, coverage expectations, and compliance requirements for TODP testing activities.

The Test Specification serves as the authoritative verification framework ensuring that all requirements, architectural decisions, technical implementations, and operational controls are validated prior to production deployment.

TEST SPEC Progress

☐ Section 1 — Document Overview & Testing Scope

☐ Section 2 — Testing Strategy & Methodology

TEST SPEC V1 — Section 2

Testing Strategy & Methodology

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

2

Version:

V1

Status:

APPROVED

2.1 Purpose

This section defines the testing strategy and methodology used throughout TODP.

The strategy establishes:

- Testing approach
- Validation methodology
- Verification workflows
- Coverage standards
- Quality gates
- Governance controls

The objective is to ensure consistent and measurable validation across all platform components.

2.2 Testing Philosophy

TODP adopts:

Risk-Based Testing

+

Requirement-Driven Testing

+

Traceability-Based Testing

All testing activities originate from approved requirements.

2.3 Verification Model

Requirement

↓

Design

↓

Implementation

↓

Test Case

↓

Execution



Verification



Approval

2.4 Testing Objectives

Verify Functional Correctness

Verify Technical Correctness

Verify Security Controls

Verify Operational Readiness

Verify Production Readiness

Verify Compliance

2.5 Testing Methodology

Planning



Design



Implementation



Execution



Analysis



Correction



Retesting



Approval

2.6 Testing Layers

Layer 1

Unit Testing

Layer 2

Integration Testing

Layer 3

API Testing

Layer 4

System Testing

Layer 5

Security Testing

Layer 6

Performance Testing

Layer 7

Acceptance Testing

2.7 Unit Testing Strategy

Purpose:

Validate Individual Components

Coverage Areas:

Functions

Classes

Services

Repositories

Utilities

Validators

2.8 Integration Testing Strategy

Purpose:

Validate Component Interactions

Coverage Areas:

Service Interactions

Database Interactions

Repository Interactions

External Integrations

Workflow Execution

2.9 API Testing Strategy

Purpose:

Validate Interface Contracts

Coverage Areas:

Endpoints

Requests

Responses

Validation Rules

Authentication

Authorization

2.10 System Testing Strategy

Purpose:

Validate Complete System Behavior

Coverage Areas:

Business Workflows

Cross-Module Operations

End-To-End Scenarios

Operational Processes

2.11 Security Testing Strategy

Purpose:

Validate Security Controls

Coverage Areas:

Authentication

Authorization

RBAC

Input Validation

Secrets

Encryption

Audit Controls

2.12 Performance Testing Strategy

Purpose:

Validate Performance Requirements

Coverage Areas:

API Latency

Database Queries

Search Performance

Ranking Performance

Recommendation Performance

System Throughput

2.13 Acceptance Testing Strategy

Purpose:

Validate Business Expectations

Coverage Areas:

Business Workflows

User Scenarios

Requirement Compliance

Governance Compliance

2.14 Requirement-Based Testing

Every requirement receives:

Requirement ID



Test Case ID



Execution Evidence



Verification Result

2.15 Risk-Based Testing

Testing priority determined by:

Business Impact

Security Impact

Operational Impact

Failure Probability

Governance Importance

2.16 Test Priority Levels

P1

Critical

P2

High

P3

Medium

P4

Low

2.17 Critical Test Areas

Authentication

Authorization

Evidence Validation

Retrieval

Ranking

Recommendation

Governance

Audit

Priority:

P1

2.18 Test Design Methodology

Identify Requirement



Identify Risks



Design Test Cases



Design Test Data



Design Expected Results



Review

2.19 Positive Testing

Purpose:

Verify Correct Behavior

Examples:

Valid Login

Valid Search

Valid Recommendation

Valid Approval

2.20 Negative Testing

Purpose:

Verify Failure Handling

Examples:

Invalid Login

Invalid Request

Invalid Policy

Invalid Data

Unauthorized Access

2.21 Boundary Testing

Purpose:

Validate Edge Conditions

Examples:

Maximum Length Inputs

Minimum Length Inputs

Empty Values

Large Datasets

2.22 Error Handling Testing

Purpose:

Validate Failure Responses

Coverage:

Validation Errors

Authorization Errors

Database Errors

System Errors

Infrastructure Errors

2.23 Regression Testing Strategy

Purpose:

Prevent Functional Regressions

Executed:

Before Releases

After Major Changes

After Defect Fixes

2.24 Smoke Testing Strategy

Purpose:

Validate Basic Platform Health

Coverage:

Authentication

Database Connectivity

Search

Ranking

Recommendation

Health Endpoints

2.25 Sanity Testing Strategy

Purpose:

Validate Recent Changes

Executed:

Post Deployment

Post Bug Fix

Post Configuration Change

2.26 Exploratory Testing Strategy

Purpose:

Discover Unexpected Issues

Focus Areas:

User Workflows

Edge Cases

Operational Workflows

Governance Workflows

2.27 Test Automation Strategy

Automation Preferred For:

Unit Tests

Integration Tests

API Tests

Regression Tests

Security Tests

2.28 Manual Testing Strategy

Manual Validation Required For:

UAT

Governance Reviews

Operational Reviews

Exploratory Testing

Compliance Validation

2.29 Test Data Strategy

Use:

Controlled Test Data

Synthetic Data

Edge Case Data

Negative Test Data

Performance Test Data

2.30 Test Environment Strategy

Separate Environments:

Development

Testing

Staging

Production

Production Testing Restricted

2.31 Entry Criteria

Testing Begins When:

Requirements Approved

Design Approved

Implementation Available

Environment Ready

Test Cases Approved

2.32 Exit Criteria

Testing Completes When:

Test Cases Executed

Critical Defects Resolved

Coverage Targets Achieved

Governance Approval Obtained

2.33 Defect Classification

Severity 1

Critical

Severity 2

High

Severity 3

Medium

Severity 4

Low

2.34 Defect Lifecycle

Reported

↓

Validated



Assigned



Fixed



Retested



Closed



2.35 Coverage Measurement

Coverage Categories:

Requirement Coverage

Code Coverage

API Coverage

Security Coverage

Workflow Coverage

2.36 Coverage Targets

Requirement Coverage:

100%

Critical Requirement Coverage:

100%

Security Coverage:

100%

Governance Coverage:

100%

2.37 Traceability Methodology

Requirement

↓

Test Case

↓

Execution

↓

Result

↓

Verification

Maintained Through:

TRM V1

2.38 Auditability Requirements

All testing activities support:

Audit Reviews

Compliance Reviews

Governance Reviews

Coverage Reviews

2.39 Reproducibility Requirements

Support:

Test Replay

Execution Replay

Result Replay

Regression Replay

2.40 Governance Rules

TEST-METH-01

Every requirement requires testing.

TEST-METH-02

Critical requirements require multiple validation methods.

TEST-METH-03

Coverage must be measurable.

TEST-METH-04

Defects must be tracked.

TEST-METH-05

Testing evidence must be retained.

TEST-METH-06

Testing must remain reproducible.

2.41 Architectural Constraints

AC-TEST-METH-01

Untested requirements are prohibited.

AC-TEST-METH-02

Critical defects cannot be ignored.

AC-TEST-METH-03

Coverage gaps require remediation.

AC-TEST-METH-04

Verification evidence cannot be deleted.

AC-TEST-METH-05

Testing cannot bypass governance.

AC-TEST-METH-06

Approval gates cannot be bypassed.

2.42 Section Summary

This section defines the testing strategy, methodology, validation framework, risk model, automation approach, defect lifecycle, coverage model, and governance controls used throughout TODP.

The methodology ensures consistent, auditable, traceable, measurable, and reproducible validation across all functional, technical, operational, security, governance, and compliance domains.

TEST SPEC Progress

▢ Section 1 — Document Overview & Testing Scope

▢ Section 2 — Testing Strategy & Methodology

▢ Section 3 — Test Environment Architecture

TEST SPEC V1 — Section 3

Test Environment Architecture

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

3

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APPROVED

3.1 Purpose

This section defines the architecture of all testing environments used throughout TODP.

The objective is to ensure:

- Consistent test execution

- Environment isolation
- Repeatable validation
- Controlled test data
- Reliable verification
- Governance compliance

3.2 Environment Principles

ENV-01

Environments must be isolated.

ENV-02

Test data must be controlled.

ENV-03

Configurations must be versioned.

ENV-04

Deployments must be reproducible.

ENV-05

Environment changes must be auditable.

ENV-06

Production isolation is mandatory.

3.3 Environment Types

Development

Testing

Staging

Production

Each environment serves a specific validation purpose.

3.4 Development Environment

Purpose:

Developer Validation

Primary Activities:

Unit Testing

Local Integration Testing

Debugging

Feature Development

3.5 Development Environment Components

Frontend

Backend

PostgreSQL

Redis

Local Observability

Mock Services

3.6 Testing Environment

Purpose:

Formal Test Execution

Primary Activities:

Integration Testing

API Testing

Security Testing

Regression Testing

3.7 Testing Environment Components

Frontend

Backend

PostgreSQL

Redis

Monitoring Stack

Logging Stack

Tracing Stack

3.8 Staging Environment

Purpose:

Production Simulation

Primary Activities:

System Testing

Performance Testing

UAT

Deployment Validation

Operational Validation

3.9 Staging Environment Components

Production-Equivalent Configuration

Production-Equivalent Infrastructure

Production-Equivalent Security

Production-Equivalent Monitoring

3.10 Production Environment

Purpose:

Live Operations

Primary Activities:

Business Operations

Monitoring

Incident Response

Operational Validation

3.11 Environment Isolation Rules

Development

≠

Testing

≠

Staging

≠

Production

Cross-environment contamination prohibited.

3.12 Configuration Management

Each Environment Maintains:

Environment Variables

Secrets

Database Configurations

Deployment Configurations

Monitoring Configurations

3.13 Environment Topology

Users



Frontend



Backend APIs



Redis



PostgreSQL



Observability Stack

3.14 Test Network Architecture

Test Traffic



Load Balancer



Application Services



Data Services



Monitoring Services

3.15 Database Architecture

Dedicated Database Per Environment

Development Database

Testing Database

Staging Database

Production Database

3.16 Redis Architecture

Dedicated Redis Per Environment

Shared Redis Instances Prohibited

3.17 Observability Architecture

Components:

Metrics

Logs

Traces

Dashboards

Alerts

3.18 Logging Environment Design

Environment-Specific Logs

Development Logs

Testing Logs

Staging Logs

Production Logs

3.19 Monitoring Environment Design

Environment-Specific Metrics

Service Metrics

Infrastructure Metrics

Application Metrics

Database Metrics

3.20 Tracing Environment Design

Distributed Tracing Enabled For:

Testing

Staging

Production

Optional In Development

3.21 Authentication Configuration

Environment-Specific Credentials

Environment-Specific Tokens

Environment-Specific Secrets

3.22 Authorization Configuration

Environment-Specific Roles

Environment-Specific Permissions

Test Roles Supported

3.23 Test Data Categories

Functional Test Data

Security Test Data

Performance Test Data

Boundary Test Data

Negative Test Data

3.24 Synthetic Data Strategy

Preferred Data Source:

Synthetic Data

Production Data Cloning Prohibited

Unless Explicitly Approved

3.25 Test Data Lifecycle

Generate



Validate



Load



Execute Tests



Reset



Archive Results

3.26 Data Reset Strategy

Support:

Database Reset

Cache Reset

Search Reset

Test Fixture Reset

3.27 Environment Provisioning

Provisioning Method:

Infrastructure As Code

Version Controlled

Auditable

3.28 Deployment Validation

Validate:

Frontend Deployment

Backend Deployment

Database Connectivity

Redis Connectivity

Monitoring Connectivity

3.29 Environment Health Checks

Required Checks:

Application Health

Database Health

Redis Health

Monitoring Health

Tracing Health

3.30 Environment Readiness Criteria

Environment Ready When:

Infrastructure Available

Services Running

Health Checks Passing

Test Data Loaded

Monitoring Active

3.31 Environment Access Controls

Access Requires:

Authentication

Authorization

Role Assignment

Audit Logging

3.32 Test Execution Isolation

Parallel Test Runs Supported

Data Collisions Prevented

Environment Integrity Preserved

3.33 Performance Test Environment

Requirements:

Production-Like Topology

Controlled Workloads

Repeatable Test Conditions

Dedicated Resources

3.34 Security Test Environment

Supports:

Authentication Testing

Authorization Testing

Penetration Testing

Vulnerability Testing

Access Control Validation

3.35 UAT Environment

Primary Users:

Product Owner

Business Reviewers

Governance Reviewers

Acceptance Reviewers

3.36 Disaster Recovery Test Environment

Supports:

Backup Testing

Restore Testing

Failover Testing

Recovery Validation

3.37 CI/CD Environment Integration

Environment Usage:

Development

↓

Testing

↓

Staging

↓

Production

Validation Required At Every Stage

3.38 Environment Auditability

Track:

Deployments

Configuration Changes

Access Events

Test Executions

Environment Changes

3.39 Environment Traceability

Maintain:

Environment Version

Deployment Version

Configuration Version

Test Version

3.40 Reproducibility Requirements

Support:

Environment Recreation

Deployment Replay

Test Replay

Validation Replay

3.41 Governance Rules

TEST-ENV-01

Environment isolation is mandatory.

TEST-ENV-02

Production data usage is restricted.

TEST-ENV-03

Environment changes require traceability.

TEST-ENV-04

Provisioning must be reproducible.

TEST-ENV-05

Access controls are mandatory.

TEST-ENV-06

Environment health validation is mandatory.

3.42 Architectural Constraints

AC-TEST-ENV-01

Environment sharing is prohibited.

AC-TEST-ENV-02

Uncontrolled test data is prohibited.

AC-TEST-ENV-03

Production environment testing is restricted.

AC-TEST-ENV-04

Environment changes cannot bypass governance.

AC-TEST-ENV-05

Health validation cannot be skipped.

AC-TEST-ENV-06

Audit logging cannot be disabled.

3.43 Section Summary

This section defines the architecture, topology, configuration strategy, provisioning model, isolation controls, test data management approach, security controls, observability integration, and governance requirements for all TODP testing environments.

The environment architecture provides the controlled foundation necessary for reliable, repeatable, auditable, and production-aligned verification activities.

TEST SPEC Progress

☐ Section 1 — Document Overview & Testing Scope

☐ Section 2 — Testing Strategy & Methodology

☐ Section 3 — Test Environment Architecture

☐ Section 4 — Unit Testing Specifications

TEST SPEC V1 — Section 4

Unit Testing Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

4

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APPROVED

4.1 Purpose

This section defines the unit testing specifications for TODP.

Unit testing validates the correctness of individual software components in isolation.

The objective is to verify:

- Business logic
- Service logic
- Repository logic
- Utility functions
- Validators
- Domain calculations

4.2 Unit Testing Objectives

UNIT-01

Verify component correctness.

UNIT-02

Detect defects early.

UNIT-03

Support regression prevention.

UNIT-04

Validate deterministic behavior.

UNIT-05

Support refactoring safety.

UNIT-06

Provide measurable coverage.

4.3 Scope

Components Covered:

Services

Repositories

Validators

DTO Mappers

Utility Functions

Calculation Engines

Policy Evaluators

4.4 Unit Testing Principles

Tests Must Be:

Independent

Repeatable

Deterministic

Fast

Auditable

Traceable

4.5 Test Structure

Arrange

↓

Act

↓

Assert

Required For Every Unit Test

4.6 Service Layer Testing

Validate:

Business Rules

Decision Logic

Validation Logic

Error Handling

State Transitions

4.7 Repository Layer Testing

Validate:

CRUD Operations

Query Logic

Filtering Logic

Pagination Logic

Constraint Handling

4.8 Validation Layer Testing

Validate:

Required Fields

Data Formats

Boundary Conditions

Invalid Inputs

Business Constraints

4.9 Utility Testing

Validate:

Helpers

Converters

Formatters

Calculators

Transformation Logic

4.10 DTO Testing

Validate:

Serialization

Deserialization

Schema Compliance

Field Mapping

Type Validation

4.11 Authentication Unit Tests

Coverage:

Password Validation

Token Generation

Token Validation

Session Validation

Credential Verification

4.12 Authorization Unit Tests

Coverage:

Role Evaluation

Permission Evaluation

Access Rules

RBAC Logic

Policy Checks

4.13 Opportunity Module Tests

Coverage:

Opportunity Creation Rules

Update Rules

Validation Rules

Version Rules

State Changes

4.14 Authority Module Tests

Coverage:

Authority Validation

Verification Logic

Confidence Calculation

Rule Enforcement

4.15 Evidence Module Tests

Coverage:

Evidence Validation

Evidence Scoring

Evidence Linking

Lineage Rules

Version Logic

4.16 Retrieval Module Tests

Coverage:

Query Processing

Search Preparation

Fusion Logic

Ranking Inputs

Result Filtering

4.17 Ranking Module Tests

Coverage:

Total Cost Formula

Weight Calculations

Ranking Logic

Tie Resolution

Version Logic

4.18 Recommendation Module Tests

Coverage:

Recommendation Logic

Explanation Generation

Decision Rules

Trace Generation

Replay Logic

4.19 Governance Module Tests

Coverage:

Policy Evaluation

Approval Rules

Compliance Logic

Governance Validation

4.20 Audit Module Tests

Coverage:

Audit Creation

Snapshot Logic

Replay Logic

Correlation Logic

4.21 Security Unit Tests

Coverage:

Authorization Checks

Input Validation

Secret Handling

Encryption Utilities

Security Rules

4.22 Positive Test Cases

Verify:

Expected Inputs

Expected Outputs

Expected State Changes

Expected Responses

4.23 Negative Test Cases

Verify:

Invalid Inputs

Unauthorized Requests

Missing Data

Malformed Data

Policy Violations

4.24 Boundary Test Cases

Verify:

Maximum Values

Minimum Values

Empty Values

Large Payloads

Edge Conditions

4.25 Exception Handling Tests

Verify:

Validation Exceptions

Business Exceptions

Repository Exceptions

Unexpected Exceptions

4.26 Mocking Strategy

Mock:

Database Calls

External Services

Cache Access

Infrastructure Dependencies

4.27 Dependency Isolation

Unit Tests Must Not Require:

Real Database

Real Network Calls

Real External Services

Production Systems

4.28 Test Data Strategy

Use:

Synthetic Test Data

Deterministic Data

Repeatable Fixtures

Controlled Inputs

4.29 Naming Convention

Format:

test_<module>_<behavior>

Examples:

test_login_success

test_ranking_formula_execution

test_policy_validation_failure

4.30 Code Coverage Targets

Minimum Coverage:

90%

Critical Modules:

95%+

4.31 Coverage Areas

Measure:

Line Coverage

Branch Coverage

Function Coverage

Class Coverage

4.32 Test Execution Frequency

Execute:

Every Commit

Every Pull Request

Every CI Build

Every Release

4.33 Defect Detection Goals

Detect:

Logic Errors

Validation Errors

Calculation Errors

Mapping Errors

Regression Defects

4.34 Pass Criteria

Unit Test Passes When:

Expected Result Returned

Expected State Achieved

No Unexpected Exceptions

Assertions Satisfied

4.35 Failure Criteria

Unit Test Fails When:

Unexpected Result

Unexpected Exception

Incorrect State

Coverage Violation

4.36 Traceability Requirements

Every Unit Test Maps To:

Requirement



Component



Test Case



Execution Result

4.37 Auditability Requirements

Maintain:

Test History

Coverage Reports

Execution Results

Defect References

4.38 Reproducibility Requirements

Support:

Test Replay

Fixture Replay

Coverage Replay

Result Replay

4.39 Governance Rules

TEST-UNIT-01

Critical logic requires unit tests.

TEST-UNIT-02

Coverage targets are mandatory.

TEST-UNIT-03

Unit tests must be automated.

TEST-UNIT-04

Unit tests must be deterministic.

TEST-UNIT-05

Test evidence must be retained.

TEST-UNIT-06

Coverage metrics must be reported.

4.40 Architectural Constraints

AC-TEST-UNIT-01

Untested business logic is prohibited.

AC-TEST-UNIT-02

External dependencies cannot be required.

AC-TEST-UNIT-03

Non-deterministic tests are prohibited.

AC-TEST-UNIT-04

Coverage gaps require remediation.

AC-TEST-UNIT-05

Critical modules cannot bypass testing.

AC-TEST-UNIT-06

Governance controls cannot be bypassed.

4.41 Section Summary

This section defines the unit testing strategy, coverage model, execution standards, isolation requirements, coverage targets, traceability controls, and governance requirements for TODP.

The unit testing layer provides the first level of verification and ensures the correctness, reliability, determinism, and maintainability of all core platform components.

TEST SPEC Progress

▢ Section 1 — Document Overview & Testing Scope

▢ Section 2 — Testing Strategy & Methodology

▢ Section 3 — Test Environment Architecture

▢ Section 4 — Unit Testing Specifications

▢ Section 5 — Integration Testing Specifications

TEST SPEC V1 — Section 5

Integration Testing Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

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5

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APPROVED

5.1 Purpose

This section defines the integration testing specifications for TODP.

Integration testing validates interactions between multiple components working together as a complete subsystem.

The objective is to verify:

- Service interactions
- Repository interactions
- Database interactions
- API-service integrations
- Cross-module workflows
- Infrastructure integrations

5.2 Integration Testing Objectives

INT-01

Verify component interoperability.

INT-02

Verify workflow execution.

INT-03

Verify data consistency.

INT-04

Verify transaction behavior.

INT-05

Verify dependency interactions.

INT-06

Verify operational stability.

5.3 Scope

Components Covered:

APIs

Services

Repositories

Database

Redis

Audit Components

Governance Components

Observability Components

5.4 Integration Testing Principles

Tests Must Be:

Repeatable

Traceable

Auditable

Deterministic

Environment Controlled

Reproducible

5.5 Integration Test Architecture

Client



API



Service



Repository



Database



Response

All layers participate in validation.

5.6 Service-to-Service Testing

Validate:

Request Flow

Response Flow

Error Handling

Dependency Behavior

Data Propagation

5.7 API-to-Service Testing

Validate:

Endpoint Routing

Input Validation

Business Logic Invocation

Response Generation

Error Responses

5.8 Service-to-Repository Testing

Validate:

Persistence Operations

Query Execution

Transaction Management

Data Retrieval

Data Updates

5.9 Repository-to-Database Testing

Validate:

Schema Compatibility

Constraint Enforcement

Index Usage

Relationship Integrity

Data Persistence

5.10 Database Transaction Testing

Validate:

Commit Operations

Rollback Operations

Atomicity

Consistency

Isolation

Durability

5.11 Authentication Integration Tests

Coverage:

Login Workflow

Session Creation

Token Refresh

Session Validation

Logout Workflow

5.12 Authorization Integration Tests

Coverage:

Role Assignment

Permission Evaluation

Access Enforcement

RBAC Execution

Governance Restrictions

5.13 Opportunity Module Tests

Coverage:

Create Opportunity Workflow

Update Workflow

Version Workflow

Search Workflow

Delete Workflow

5.14 Authority Module Tests

Coverage:

Authority Creation

Authority Verification

Confidence Scoring

Authority Retrieval

5.15 Evidence Module Tests

Coverage:

Evidence Creation

Validation Workflow

Version Management

Lineage Workflow

Evidence Retrieval

5.16 Retrieval Module Tests

Coverage:

Search Workflow

BM25 Retrieval

Vector Retrieval

Hybrid Retrieval

Result Aggregation

5.17 Ranking Module Tests

Coverage:

Ranking Execution

Formula Processing

Ranking Storage

Replay Workflow

Version Selection

5.18 Recommendation Module Tests

Coverage:

Recommendation Generation

Explanation Creation

Traceability Generation

Replay Execution

Recommendation Storage

5.19 Governance Module Tests

Coverage:

Policy Validation

Approval Workflow

Governance Enforcement

Compliance Validation

5.20 Audit Module Tests

Coverage:

Audit Record Creation

Correlation Tracking

Snapshot Creation

Replay Workflow

Audit Retrieval

5.21 Redis Integration Tests

Validate:

Cache Writes

Cache Reads

Cache Expiration

Key Management

Fallback Logic

5.22 Observability Integration Tests

Validate:

Log Generation

Metric Collection

Trace Creation

Alert Creation

Correlation Tracking

5.23 Security Integration Tests

Validate:

Authentication Enforcement

Authorization Enforcement

Input Validation

Audit Generation

Access Controls

5.24 Positive Workflow Tests

Verify:

Successful Execution Paths

Expected State Changes

Expected Persistence

Expected Responses

5.25 Negative Workflow Tests

Verify:

Invalid Requests

Missing Data

Unauthorized Access

Policy Violations

Workflow Failures

5.26 Error Recovery Tests

Validate:

Database Failures

Cache Failures

Validation Failures

Service Failures

Transaction Failures

5.27 Cross-Module Testing

Validate:

Opportunity ↔ Evidence

Authority ↔ Evidence

Retrieval ↔ Ranking

Ranking ↔ Recommendation

Governance ↔ Approval

Audit ↔ All Modules

5.28 Data Integrity Testing

Validate:

Relationship Integrity

Referential Integrity

Version Integrity

Audit Integrity

Governance Integrity

5.29 Test Data Strategy

Use:

Controlled Fixtures

Synthetic Data

Workflow Data

Boundary Data

Negative Data

5.30 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Production Execution Restricted

5.31 Test Execution Frequency

Execute:

Every Pull Request

Nightly Builds

Release Candidates

Pre-Production Validation

5.32 Coverage Targets

Workflow Coverage:

100%

Critical Workflow Coverage:

100%

Integration Coverage:

95%+

5.33 Performance Expectations

Validate:

Workflow Completion

Response Consistency

Data Consistency

Operational Stability

5.34 Defect Detection Goals

Detect:

Integration Defects

Transaction Defects

Data Flow Defects

Dependency Defects

Workflow Defects

5.35 Pass Criteria

Integration Test Passes When:

Workflow Completes

Data Persists Correctly

Responses Are Correct

No Unexpected Failures

5.36 Failure Criteria

Integration Test Fails When:

Workflow Breaks

Data Corrupts

Transaction Fails

Unexpected Response Occurs

5.37 Traceability Requirements

Every Integration Test Maps To:

Requirement

↓

Workflow

↓

Implementation



Execution Result

5.38 Auditability Requirements

Maintain:

Execution Logs

Coverage Reports

Defect Reports

Validation Evidence

5.39 Reproducibility Requirements

Support:

Workflow Replay

Execution Replay

Environment Replay

Result Replay

5.40 Governance Rules

TEST-INT-01

Critical workflows require integration testing.

TEST-INT-02

Cross-module interactions require validation.

TEST-INT-03

Data integrity must be verified.

TEST-INT-04

Workflow coverage must be measurable.

TEST-INT-05

Execution evidence must be retained.

TEST-INT-06

Integration tests must be reproducible.

5.41 Architectural Constraints

AC-TEST-INT-01

Critical workflows cannot bypass integration testing.

AC-TEST-INT-02

Data integrity validation is mandatory.

AC-TEST-INT-03

Cross-module testing cannot be skipped.

AC-TEST-INT-04

Coverage gaps require remediation.

AC-TEST-INT-05

Verification evidence cannot be deleted.

AC-TEST-INT-06

Governance controls cannot be bypassed.

5.42 Section Summary

This section defines the integration testing strategy, workflow validation model, transaction verification approach, cross-module testing requirements, coverage targets, traceability controls, and governance requirements for TODP.

The integration testing layer ensures that independently validated components operate correctly together and support reliable execution of all business, technical, governance, audit, and operational workflows.

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TEST SPEC V1 — Section 6

API Testing Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

6

Version:

V1

Status:

APPROVED

6.1 Purpose

This section defines the API testing specifications for TODP.

API testing validates that all exposed interfaces operate according to approved contracts defined in IAD V1 and implemented in TDD V1.

The objective is to verify:

- Endpoint behavior
- Request validation
- Response validation
- Authentication enforcement
- Authorization enforcement
- Error handling

6.2 API Testing Objectives

API-01

Verify endpoint correctness.

API-02

Verify contract compliance.

API-03

Verify validation rules.

API-04

Verify security controls.

API-05

Verify response consistency.

API-06

Verify interoperability.

6.3 Scope

Covered APIs:

Authentication

Authorization

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Operational APIs

6.4 API Testing Principles

Tests Must Be:

Repeatable

Deterministic

Auditable

Traceable

Automated

Reproducible

6.5 API Validation Model

Request



Validation



Authentication



Authorization



Business Logic



Response



Verification

6.6 Contract Validation

Verify:

Request Schema

Response Schema

Field Types

Required Fields

Optional Fields

Validation Rules

6.7 HTTP Method Validation

Verify:

GET

POST

PUT

PATCH

DELETE

Correct Method Usage Required

6.8 Status Code Validation

Verify:

200 OK

201 Created

204 No Content

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found

409 Conflict

422 Validation Error

500 Internal Server Error

6.9 Authentication API Tests

Coverage:

Login

Logout

Refresh Token

Session Validation

Password Management

6.10 Authorization API Tests

Coverage:

Role APIs

Permission APIs

RBAC Enforcement

Access Restrictions

Governance Restrictions

6.11 Opportunity API Tests

Coverage:

Create Opportunity

Update Opportunity

Delete Opportunity

Search Opportunity

Version Retrieval

6.12 Authority API Tests

Coverage:

Create Authority

Verify Authority

Update Authority

Retrieve Authority

Confidence Retrieval

6.13 Evidence API Tests

Coverage:

Create Evidence

Validate Evidence

Update Evidence

Retrieve Evidence

Version Retrieval

Lineage Retrieval

6.14 Retrieval API Tests

Coverage:

Keyword Search

Semantic Search

Hybrid Search

Search Replay

Search History

6.15 Ranking API Tests

Coverage:

Execute Ranking

Retrieve Rankings

Formula Retrieval

Replay Ranking

Version Retrieval

6.16 Recommendation API Tests

Coverage:

Generate Recommendation

Retrieve Recommendation

Retrieve Explanation

Retrieve Trace

Replay Recommendation

6.17 Governance API Tests

Coverage:

Policy Validation

Approval Workflow

Governance Retrieval

Compliance Reporting

6.18 Audit API Tests

Coverage:

Audit Retrieval

Snapshot Retrieval

Replay Execution

Correlation Retrieval

Audit Filtering

6.19 Operational API Tests

Coverage:

Health Endpoints

Readiness Endpoints

Liveness Endpoints

Monitoring Endpoints

6.20 Positive API Tests

Verify:

Valid Requests

Valid Inputs

Valid Authorization

Expected Responses

Expected Persistence

6.21 Negative API Tests

Verify:

Invalid Inputs

Invalid Payloads

Missing Fields

Unauthorized Requests

Forbidden Requests

6.22 Boundary API Tests

Verify:

Maximum Field Lengths

Minimum Field Lengths

Large Payloads

Empty Values

Edge Conditions

6.23 Response Validation Tests

Verify:

Response Structure

Response Types

Response Metadata

Response Consistency

Response Completeness

6.24 Error Handling Tests

Verify:

Validation Errors

Authentication Errors

Authorization Errors

Business Rule Errors

Server Errors

6.25 Pagination Tests

Verify:

Page Size

Page Number

Sorting

Filtering

Navigation Metadata

6.26 Search API Tests

Verify:

Search Accuracy

Filter Accuracy

Sorting Accuracy

Pagination Accuracy

Replay Accuracy

6.27 Security API Tests

Verify:

JWT Validation

RBAC Enforcement

Input Sanitization

Rate Limiting

Audit Generation

6.28 Performance API Tests

Verify:

Response Time

Throughput

Concurrency

Latency

Error Rates

6.29 API Versioning Tests

Verify:

Version Availability

Backward Compatibility

Contract Stability

Version Routing

6.30 Idempotency Tests

Verify:

Repeated Requests

Duplicate Requests

Safe Retries

Consistent Outcomes

6.31 Data Consistency Tests

Verify:

Create Operations

Update Operations

Delete Operations

Read Operations

Transaction Integrity

6.32 API Workflow Tests

Validate:

Authentication Flow

Opportunity Flow

Evidence Flow

Retrieval Flow

Ranking Flow

Recommendation Flow

Governance Flow

Audit Flow

6.33 Automation Requirements

API Tests Must Be:

Automated

Version Controlled

Repeatable

CI/CD Integrated

6.34 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Production Testing Restricted

6.35 Coverage Targets

Endpoint Coverage:

100%

Critical Endpoint Coverage:

100%

Contract Coverage:

100%

6.36 Defect Detection Goals

Detect:

Contract Violations

Validation Defects

Security Defects

Routing Defects

Workflow Defects

6.37 Pass Criteria

API Test Passes When:

Correct Status Returned

Correct Response Returned

Contract Satisfied

Security Controls Enforced

6.38 Failure Criteria

API Test Fails When:

Incorrect Response

Incorrect Status

Contract Violation

Security Violation

Unexpected Behavior

6.39 Traceability Requirements

Every API Test Maps To:

Requirement



API Contract



Endpoint



Execution Result

6.40 Auditability Requirements

Maintain:

Execution Reports

Coverage Reports

Contract Reports

Defect Reports

6.41 Reproducibility Requirements

Support:

API Replay

Request Replay

Response Replay

Execution Replay

6.42 Governance Rules

TEST-API-01

Every endpoint requires API testing.

TEST-API-02

Contract validation is mandatory.

TEST-API-03

Security validation is mandatory.

TEST-API-04

Coverage must be measurable.

TEST-API-05

Execution evidence must be retained.

TEST-API-06

API tests must be reproducible.

6.43 Architectural Constraints

AC-TEST-API-01

Unvalidated APIs are prohibited.

AC-TEST-API-02

Contract violations cannot be ignored.

AC-TEST-API-03

Security validation cannot be skipped.

AC-TEST-API-04

Coverage gaps require remediation.

AC-TEST-API-05

Verification evidence cannot be deleted.

AC-TEST-API-06

Governance controls cannot be bypassed.

6.44 Section Summary

This section defines the API testing strategy, contract validation model, endpoint coverage requirements, security verification approach, automation standards, traceability controls, and governance requirements for TODP.

The API testing layer ensures that all externally exposed interfaces operate consistently, securely, predictably, and in compliance with approved interface contracts and business requirements.

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TEST SPEC V1 — Section 7

Database Testing Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

7

Version:

V1

Status:

APPROVED

7.1 Purpose

This section defines the database testing specifications for TODP.

Database testing validates the correctness, integrity, consistency, security, and performance of the persistence layer.

The objective is to verify:

- Data integrity
- Schema correctness
- Relationship integrity
- Transaction behavior
- Migration reliability
- Persistence consistency

7.2 Database Testing Objectives

DB-01

Verify schema correctness.

DB-02

Verify data integrity.

DB-03

Verify relationship integrity.

DB-04

Verify transaction behavior.

DB-05

Verify migration reliability.

DB-06

Verify persistence consistency.

7.3 Scope

Components Covered:

PostgreSQL

Database Schemas

Tables

Indexes

Constraints

Views

Audit Tables

Version Tables

7.4 Database Testing Principles

Tests Must Be:

Repeatable

Auditable

Traceable

Deterministic

Reproducible

Environment Controlled

7.5 Database Validation Model

Schema



Table



Constraint



Relationship



Transaction



Verification

7.6 Schema Validation

Verify:

Table Creation

Column Definitions

Data Types

Defaults

Nullability

Schema Consistency

7.7 Table Validation

Verify:

Table Structure

Column Names

Column Types

Required Fields

Optional Fields

Metadata Fields

7.8 Primary Key Validation

Verify:

Uniqueness

Non-Null Values

Key Enforcement

Entity Identity

7.9 Foreign Key Validation

Verify:

Referential Integrity

Relationship Enforcement

Delete Rules

Update Rules

Constraint Enforcement

7.10 Unique Constraint Validation

Verify:

Duplicate Prevention

Constraint Enforcement

Error Handling

Data Consistency

7.11 Check Constraint Validation

Verify:

Business Rules

Data Boundaries

Value Restrictions

Validation Logic

7.12 Index Validation

Verify:

Index Creation

Index Usage

Query Optimization

Search Efficiency

7.13 Migration Testing

Coverage:

Migration Execution

Migration Rollback

Migration Replay

Version Consistency

Schema Consistency

7.14 Alembic Validation

Verify:

Upgrade Scripts

Downgrade Scripts

Version Tracking

Migration Ordering

7.15 Authentication Database Tests

Validate:

Users Table

Sessions Table

Refresh Tokens

Credential Metadata

Relationship Integrity

7.16 Authorization Database Tests

Validate:

Roles Table

Permissions Table

Role Permission Mapping

Access Integrity

7.17 Opportunity Database Tests

Validate:

Opportunity Storage

Version Storage

History Tracking

Search Metadata

7.18 Authority Database Tests

Validate:

Authority Storage

Verification Records

Confidence Scores

Relationship Integrity

7.19 Evidence Database Tests

Validate:

Evidence Storage

Validation Records

Version Records

Lineage Records

7.20 Retrieval Database Tests

Validate:

Search Records

Embeddings

Search History

Retrieval Results

Replay Records

7.21 Ranking Database Tests

Validate:

Ranking Executions

Formula Versions

Ranking Results

Replay Records

7.22 Recommendation Database Tests

Validate:

Recommendations

Explanations

Traceability Links

History Records

7.23 Governance Database Tests

Validate:

Policies

Approvals

Governance Records

Compliance Records

7.24 Audit Database Tests

Validate:

Audit Records

Snapshots

Correlations

Replay Records

7.25 Transaction Testing

Verify:

Commit Behavior

Rollback Behavior

Isolation Behavior

Consistency Behavior

Atomicity

7.26 Concurrency Testing

Verify:

Concurrent Reads

Concurrent Writes

Lock Handling

Transaction Integrity

Conflict Resolution

7.27 Data Integrity Testing

Verify:

Referential Integrity

Version Integrity

Audit Integrity

Governance Integrity

Relationship Integrity

7.28 CRUD Testing

Validate:

Create

Read

Update

Delete

Soft Delete

Recovery Operations

7.29 Audit Trail Testing

Verify:

Audit Creation

Audit Updates

Snapshot Creation

Correlation Tracking

Replay Data

7.30 Data Retention Testing

Verify:

Retention Policies

Archival Policies

Audit Retention

Governance Retention

7.31 Backup Testing

Validate:

Backup Creation

Backup Integrity

Backup Storage

Backup Recovery

7.32 Restore Testing

Validate:

Database Restore

Schema Restore

Data Restore

Audit Restore

Version Restore

7.33 Performance Testing

Verify:

Query Performance

Index Performance

Write Performance

Read Performance

Transaction Performance

7.34 Security Testing

Verify:

Access Restrictions

Database Permissions

Encryption Support

Audit Logging

Secret Protection

7.35 Test Data Validation

Use:

Synthetic Data

Boundary Data

Negative Data

Performance Data

Relationship Data

7.36 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Production Testing Restricted

7.37 Coverage Targets

Schema Coverage:

100%

Constraint Coverage:

100%

Migration Coverage:

100%

Critical Table Coverage:

100%

7.38 Defect Detection Goals

Detect:

Schema Defects

Relationship Defects

Migration Defects

Constraint Defects

Transaction Defects

Performance Defects

7.39 Pass Criteria

Database Test Passes When:

Schema Valid

Relationships Valid

Constraints Enforced

Transactions Correct

Performance Acceptable

7.40 Failure Criteria

Database Test Fails When:

Constraint Violations

Relationship Failures

Migration Failures

Data Corruption

Unexpected Behavior

7.41 Traceability Requirements

Every Database Test Maps To:

Requirement



Entity



Table



Execution Result

7.42 Auditability Requirements

Maintain:

Migration Reports

Execution Reports

Coverage Reports

Defect Reports

Integrity Reports

7.43 Reproducibility Requirements

Support:

Migration Replay

Schema Replay

Backup Replay

Restore Replay

Execution Replay

7.44 Governance Rules

TEST-DB-01

Every persistent entity requires validation.

TEST-DB-02

Migration validation is mandatory.

TEST-DB-03

Data integrity validation is mandatory.

TEST-DB-04

Coverage must be measurable.

TEST-DB-05

Execution evidence must be retained.

TEST-DB-06

Database tests must be reproducible.

7.45 Architectural Constraints

AC-TEST-DB-01

Unvalidated schemas are prohibited.

AC-TEST-DB-02

Migration testing cannot be skipped.

AC-TEST-DB-03

Integrity validation is mandatory.

AC-TEST-DB-04

Coverage gaps require remediation.

AC-TEST-DB-05

Verification evidence cannot be deleted.

AC-TEST-DB-06

Governance controls cannot be bypassed.

7.46 Section Summary

This section defines the database testing strategy, schema validation model, migration verification approach, integrity testing requirements, backup and recovery validation standards, traceability controls, and governance requirements for TODP.

The database testing layer ensures that all persistence mechanisms operate reliably, securely, consistently, and in compliance with approved data architecture and governance requirements.

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TEST SPEC V1 — Section 8

Authentication Testing Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

8

Version:

V1

Status:

APPROVED

8.1 Purpose

This section defines the authentication testing specifications for TODP.

Authentication testing validates identity verification mechanisms and ensures that only legitimate users can access protected platform resources.

The objective is to verify:

- Identity verification
- Session management
- Token management

- Credential security
- Access protection
- Authentication auditability

8.2 Authentication Testing Objectives

AUTH-TEST-01

Verify user authentication.

AUTH-TEST-02

Verify credential validation.

AUTH-TEST-03

Verify session security.

AUTH-TEST-04

Verify token security.

AUTH-TEST-05

Verify authentication auditability.

AUTH-TEST-06

Verify authentication resilience.

8.3 Scope

Components Covered:

Login APIs

Logout APIs

Session Management

JWT Tokens

Refresh Tokens

Password Management

Credential Validation

Authentication Middleware

8.4 Authentication Validation Model

Credential



Authentication Request



Identity Verification



Session Creation



Token Issuance



Authorization Access

8.5 Login Testing

Validate:

Valid Login

Invalid Login

Locked Account Login

Disabled Account Login

Expired Credential Login

8.6 Successful Login Tests

Verify:

Correct Credentials

Session Creation

Token Creation

Audit Record Creation

Expected Response

8.7 Failed Login Tests

Verify:

Incorrect Password

Unknown User

Disabled User

Expired User

Malformed Request

8.8 Password Validation Tests

Verify:

Minimum Length

Maximum Length

Complexity Rules

Restricted Patterns

Password Reuse Prevention

8.9 Password Change Tests

Verify:

Valid Password Change

Invalid Current Password

Weak Password Rejection

Password Policy Enforcement

Audit Generation

8.10 Session Management Tests

Validate:

Session Creation

Session Validation

Session Expiration

Session Revocation

Session Cleanup

8.11 Session Timeout Tests

Verify:

Idle Timeout

Maximum Lifetime

Forced Expiration

Expired Session Rejection

8.12 JWT Token Tests

Verify:

Token Creation

Token Validation

Token Expiration

Token Claims

Signature Validation

8.13 Refresh Token Tests

Verify:

Refresh Generation

Refresh Validation

Refresh Rotation

Refresh Expiration

Refresh Revocation

8.14 Token Replay Tests

Verify:

Replay Prevention

Duplicate Usage Detection

Revoked Token Rejection

Expired Token Rejection

8.15 Authentication Middleware Tests

Validate:

Authenticated Requests

Unauthenticated Requests

Token Validation

Identity Injection

Request Rejection

8.16 Account Status Tests

Verify:

Active Accounts

Disabled Accounts

Locked Accounts

Expired Accounts

Suspended Accounts

8.17 Logout Tests

Validate:

Session Revocation

Token Invalidation

Refresh Token Revocation

Audit Generation

8.18 Multi-Session Tests

Verify:

Multiple Active Sessions

Session Isolation

Session Revocation

Concurrent Access Handling

8.19 Credential Storage Tests

Verify:

Hash Storage

Hash Verification

No Plaintext Storage

Credential Protection

8.20 Security Rule Tests

Verify:

Brute Force Protection

Credential Validation

Token Security

Session Security

Authentication Enforcement

8.21 Authentication Audit Tests

Verify:

Login Events

Logout Events

Failed Login Events

Password Changes

Session Events

8.22 Positive Authentication Tests

Validate:

Valid Credentials

Valid Tokens

Valid Sessions

Valid Requests

8.23 Negative Authentication Tests

Validate:

Invalid Credentials

Expired Tokens

Revoked Tokens

Invalid Sessions

Malformed Requests

8.24 Boundary Authentication Tests

Verify:

Maximum Password Length

Minimum Password Length

Maximum Session Lifetime

Maximum Token Lifetime

8.25 Error Handling Tests

Validate:

Authentication Failures

Token Failures

Session Failures

Credential Failures

Unexpected Errors

8.26 Concurrent Authentication Tests

Verify:

Concurrent Logins

Concurrent Refresh Requests

Concurrent Session Validation

Concurrent Logout Operations

8.27 Performance Tests

Validate:

Login Latency

Token Validation Latency

Session Validation Latency

Concurrent Authentication Requests

8.28 Resilience Tests

Verify:

Database Failure Handling

Redis Failure Handling

Session Recovery

Graceful Degradation

8.29 Security Attack Simulations

Validate Resistance Against:

Credential Stuffing

Brute Force Attempts

Token Replay

Session Hijacking Attempts

Unauthorized Access

8.30 Authentication Workflow Tests

Validate End-To-End Flow:

Login



Session Creation



Token Issuance



Authorized Access



Logout



Revocation

8.31 Test Data Requirements

Use:

Valid Credentials

Invalid Credentials

Expired Credentials

Disabled Accounts

Synthetic Users

8.32 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Security Test Environment

8.33 Coverage Targets

Authentication Coverage:

100%

Critical Authentication Coverage:

100%

Security Coverage:

100%

8.34 Defect Detection Goals

Detect:

Authentication Failures

Session Defects

Token Defects

Credential Defects

Security Defects

Audit Defects

8.35 Pass Criteria

Authentication Test Passes When:

Identity Verified Correctly

Security Rules Enforced

Tokens Behave Correctly

Sessions Behave Correctly

Audit Records Created

8.36 Failure Criteria

Authentication Test Fails When:

Unauthorized Access Granted

Token Validation Fails

Session Validation Fails

Credential Rules Fail

Audit Records Missing

8.37 Traceability Requirements

Every Authentication Test Maps To:

Requirement



Authentication Component



Execution Result



Verification Record

8.38 Auditability Requirements

Maintain:

Execution Reports

Authentication Logs

Coverage Reports

Security Reports

Defect Reports

8.39 Reproducibility Requirements

Support:

Login Replay

Token Replay Validation

Session Replay

Execution Replay

Result Replay

8.40 Governance Rules

TEST-AUTH-01

All authentication mechanisms require testing.

TEST-AUTH-02

Token validation testing is mandatory.

TEST-AUTH-03

Session validation testing is mandatory.

TEST-AUTH-04

Coverage must be measurable.

TEST-AUTH-05

Security validation is mandatory.

TEST-AUTH-06

Authentication tests must be reproducible.

8.41 Architectural Constraints

AC-TEST-AUTH-01

Untested authentication paths are prohibited.

AC-TEST-AUTH-02

Token validation cannot be bypassed.

AC-TEST-AUTH-03

Session validation cannot be bypassed.

AC-TEST-AUTH-04

Coverage gaps require remediation.

AC-TEST-AUTH-05

Verification evidence cannot be deleted.

AC-TEST-AUTH-06

Governance controls cannot be bypassed.

8.42 Section Summary

This section defines the authentication testing strategy, session validation approach, token verification model, credential security validation requirements, traceability controls, and governance requirements for TODP.

The authentication testing layer ensures that identity verification mechanisms remain secure, reliable, auditable, resilient, and compliant with platform security requirements.

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▢ Section 8 — Authentication Testing Specifications

▢ Section 9 — Authorization & RBAC Testing Specifications

TEST SPEC V1 — Section 9

Authorization & RBAC Testing Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

9

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V1

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APPROVED

9.1 Purpose

This section defines the authorization and Role-Based Access Control (RBAC) testing specifications for TODP.

Authorization testing validates that authenticated users can only access resources and operations permitted by their assigned roles and permissions.

The objective is to verify:

- Access control enforcement
- Role assignment behavior
- Permission evaluation
- RBAC policy execution
- Governance restrictions
- Auditability of access decisions

9.2 Authorization Testing Objectives

AUTHZ-TEST-01

Verify access control enforcement.

AUTHZ-TEST-02

Verify role evaluation.

AUTHZ-TEST-03

Verify permission evaluation.

AUTHZ-TEST-04

Verify governance restrictions.

AUTHZ-TEST-05

Verify audit generation.

AUTHZ-TEST-06

Verify authorization resilience.

9.3 Scope

Components Covered:

RBAC Engine

Role Management

Permission Management

Authorization Middleware

Policy Evaluators

Governance Controls

Audit Logging

Protected APIs

9.4 Authorization Validation Model

Authenticated User

↓

Role Resolution

↓

Permission Resolution

↓

Policy Evaluation

↓

Access Decision

↓

Audit Record

9.5 Role Validation Tests

Verify:

Role Assignment

Role Removal

Role Updates

Multiple Roles

Role Inheritance

9.6 Permission Validation Tests

Verify:

Permission Assignment

Permission Revocation

Permission Updates

Permission Resolution

Permission Enforcement

9.7 RBAC Evaluation Tests

Validate:

Role Lookup

Permission Lookup

Access Evaluation

Decision Consistency

Policy Enforcement

9.8 Access Control Tests

Verify:

Authorized Access

Unauthorized Access

Restricted Operations

Resource Protection

Operation Protection

9.9 API Authorization Tests

Validate:

Protected Endpoints

Role-Based Endpoints

Permission-Based Endpoints

Governance Endpoints

Administrative Endpoints

9.10 Administrative Access Tests

Verify:

Admin Access

Admin Restrictions

Privilege Enforcement

Administrative Audit Trails

9.11 Governance Access Tests

Validate:

Policy Management Access

Approval Access

Compliance Access

Audit Access

Governance Restrictions

9.12 Opportunity Authorization Tests

Verify:

Opportunity Creation Access

Opportunity Update Access

Opportunity Delete Access

Opportunity View Access

9.13 Authority Authorization Tests

Verify:

Authority Management Access

Verification Access

Confidence Management Access

Authority Review Access

9.14 Evidence Authorization Tests

Verify:

Evidence Creation Access

Evidence Validation Access

Evidence Modification Access

Evidence Review Access

9.15 Retrieval Authorization Tests

Verify:

Search Execution Access

Search Replay Access

Search History Access

Administrative Search Access

9.16 Ranking Authorization Tests

Verify:

Ranking Execution Access

Formula Management Access

Replay Access

Ranking Review Access

9.17 Recommendation Authorization Tests

Verify:

Recommendation Generation Access

Recommendation Review Access

Explanation Access

Replay Access

9.18 Audit Authorization Tests

Verify:

Audit Access

Replay Access

Snapshot Access

Correlation Access

Administrative Restrictions

9.19 Policy Evaluation Tests

Validate:

Allow Rules

Deny Rules

Conditional Rules

Conflict Resolution

Policy Priority

9.20 Least Privilege Tests

Verify:

Minimum Required Access

Excess Privilege Prevention

Role Boundary Enforcement

Permission Boundary Enforcement

9.21 Segregation Of Duties Tests

Validate:

Conflicting Role Detection

Restricted Combinations

Governance Separation

Approval Separation

Audit Separation

9.22 Role Escalation Tests

Verify:

Privilege Escalation Prevention

Unauthorized Role Changes

Unauthorized Permission Changes

Administrative Controls

9.23 Access Revocation Tests

Validate:

Role Revocation

Permission Revocation

Immediate Enforcement

Session Revalidation

9.24 Authorization Middleware Tests

Verify:

Access Evaluation

Decision Enforcement

Error Handling

Audit Generation

Request Rejection

9.25 Positive Authorization Tests

Validate:

Allowed Access

Valid Permissions

Valid Roles

Approved Actions

Expected Results

9.26 Negative Authorization Tests

Validate:

Denied Access

Missing Permissions

Missing Roles

Restricted Operations

Policy Violations

9.27 Boundary Authorization Tests

Verify:

Maximum Roles

Maximum Permissions

Complex Role Combinations

Large Permission Sets

9.28 Concurrent Authorization Tests

Validate:

Concurrent Access Requests

Concurrent Permission Updates

Concurrent Role Changes

Decision Consistency

9.29 Security Attack Simulations

Validate Resistance Against:

Privilege Escalation

Permission Bypass

Role Manipulation

Unauthorized Access

Policy Circumvention

9.30 Authorization Audit Tests

Verify:

Access Decisions Logged

Permission Changes Logged

Role Changes Logged

Administrative Actions Logged

Policy Changes Logged

9.31 Access Decision Traceability

Verify:

Decision Source

Role Source

Permission Source

Policy Source

Audit Record

9.32 Performance Tests

Validate:

Authorization Latency

Permission Resolution Time

Role Resolution Time

Concurrent Decision Processing

9.33 Resilience Tests

Verify:

Database Failures

Permission Store Failures

Role Store Failures

Graceful Failure Handling

9.34 Test Data Requirements

Use:

Valid Roles

Invalid Roles

Valid Permissions

Invalid Permissions

Synthetic Users

Synthetic Policies

9.35 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Security Test Environment

9.36 Coverage Targets

Authorization Coverage:

100%

RBAC Coverage:

100%

Critical Access Coverage:

100%

Governance Coverage:

100%

9.37 Defect Detection Goals

Detect:

Access Control Defects

RBAC Defects

Permission Defects

Policy Defects

Governance Defects

Audit Defects

9.38 Pass Criteria

Authorization Test Passes When:

Correct Access Decision Returned

Policies Enforced Correctly

Permissions Evaluated Correctly

Audit Records Generated

No Unauthorized Access Allowed

9.39 Failure Criteria

Authorization Test Fails When:

Unauthorized Access Granted

Authorized Access Denied

Policy Evaluation Incorrect

Audit Records Missing

Privilege Escalation Possible

9.40 Traceability Requirements

Every Authorization Test Maps To:

Requirement



Role



Permission



Policy



Execution Result

9.41 Auditability Requirements

Maintain:

Access Logs

Authorization Reports

Coverage Reports

Defect Reports

Policy Validation Reports

9.42 Reproducibility Requirements

Support:

Access Replay

Policy Replay

Decision Replay

Execution Replay

Result Replay

9.43 Governance Rules

TEST-AUTHZ-01

All protected resources require authorization testing.

TEST-AUTHZ-02

RBAC validation is mandatory.

TEST-AUTHZ-03

Policy validation is mandatory.

TEST-AUTHZ-04

Coverage must be measurable.

TEST-AUTHZ-05

Audit generation validation is mandatory.

TEST-AUTHZ-06

Authorization tests must be reproducible.

9.44 Architectural Constraints

AC-TEST-AUTHZ-01

Untested access paths are prohibited.

AC-TEST-AUTHZ-02

RBAC validation cannot be skipped.

AC-TEST-AUTHZ-03

Privilege escalation risks must be remediated.

AC-TEST-AUTHZ-04

Coverage gaps require remediation.

AC-TEST-AUTHZ-05

Verification evidence cannot be deleted.

AC-TEST-AUTHZ-06

Governance controls cannot be bypassed.

9.45 Section Summary

This section defines the authorization and RBAC testing strategy, access control validation approach, policy evaluation requirements, privilege management controls, traceability standards, and governance requirements for TODP.

The authorization testing layer ensures that all platform resources are protected through consistent, auditable, secure, and governance-compliant access control mechanisms.

TEST SPEC Progress

- 🔗 Section 1 — Document Overview & Testing Scope
- 🔗 Section 2 — Testing Strategy & Methodology
- 🔗 Section 3 — Test Environment Architecture
- 🔗 Section 4 — Unit Testing Specifications
- 🔗 Section 5 — Integration Testing Specifications
- 🔗 Section 6 — API Testing Specifications
- 🔗 Section 7 — Database Testing Specifications
- 🔗 Section 8 — Authentication Testing Specifications
- 🔗 Section 9 — Authorization & RBAC Testing Specifications

- 🔗 Section 10 — Opportunity Module Test Specifications

TEST SPEC V1 — Section 10

Opportunity Module Test Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

10

Version:

V1

Status:

APPROVED

10.1 Purpose

This section defines the testing specifications for the Opportunity Module.

The Opportunity Module is responsible for creation, management, lifecycle control, searchability, versioning, governance tracking, and auditability of opportunities within TODP.

The objective is to verify:

- Opportunity lifecycle correctness
- Data integrity
- Validation rules
- Search behavior
- Version management
- Governance compliance

10.2 Testing Objectives

OPP-TEST-01

Verify opportunity creation.

OPP-TEST-02

Verify opportunity updates.

OPP-TEST-03

Verify opportunity retrieval.

OPP-TEST-04

Verify opportunity versioning.

OPP-TEST-05

Verify governance controls.

OPP-TEST-06

Verify auditability.

10.3 Scope

Components Covered:

Opportunity APIs

Opportunity Services

Opportunity Repositories

Opportunity Database Tables

Opportunity Validators

Opportunity Audit Records

Opportunity Search Metadata

10.4 Opportunity Lifecycle Validation

Opportunity Draft



Validation



Approval



Active



Archived



Historical Record

All state transitions must be validated.

10.5 Opportunity Creation Tests

Verify:

Valid Creation

Required Fields

Optional Fields

Metadata Creation

Audit Creation

Version Creation

10.6 Opportunity Creation Negative Tests

Verify:

Missing Required Fields

Invalid Field Types

Duplicate Requests

Malformed Requests

Unauthorized Requests

10.7 Opportunity Update Tests

Validate:

Valid Updates

Metadata Updates

Version Creation

Audit Updates

History Tracking

10.8 Opportunity Update Negative Tests

Verify:

Unauthorized Updates

Invalid Data

Archived Opportunity Updates

Policy Violations

Version Conflicts

10.9 Opportunity Retrieval Tests

Verify:

Retrieve By ID

Retrieve By Filters

Retrieve By Status

Retrieve By Metadata

Pagination Support

10.10 Opportunity Search Tests

Validate:

Keyword Search

Filter Search

Status Search

Metadata Search

Combined Filters

10.11 Opportunity Search Accuracy Tests

Verify:

Expected Results Returned

Incorrect Results Excluded

Sorting Accuracy

Pagination Accuracy

Search Consistency

10.12 Opportunity Status Tests

Validate:

Draft Status

Approved Status

Active Status

Archived Status

Historical Status

10.13 Opportunity State Transition Tests

Verify:

Draft → Approved

Approved → Active

Active → Archived

Historical Record Creation

Invalid Transition Rejection

10.14 Opportunity Versioning Tests

Validate:

Version Creation

Version Retrieval

Version Comparison

Version Replay

Version Integrity

10.15 Opportunity History Tests

Verify:

History Creation

History Retrieval

History Accuracy

Timeline Integrity

Audit Consistency

10.16 Opportunity Ownership Tests

Validate:

Owner Assignment

Owner Changes

Ownership Restrictions

Ownership Audit Records

10.17 Opportunity Validation Rules

Verify:

Required Attributes

Format Validation

Business Constraints

Governance Constraints

Data Consistency

10.18 Opportunity Relationship Tests

Validate:

Opportunity ↔ Evidence

Opportunity ↔ Authority

Opportunity ↔ Recommendations

Opportunity ↔ Audit Records

10.19 Opportunity Governance Tests

Verify:

Approval Requirements

Policy Enforcement

Governance Restrictions

Compliance Validation

Approval Traceability

10.20 Opportunity Audit Tests

Validate:

Create Audit Records

Update Audit Records

Status Change Audits

Ownership Audits

Version Audits

10.21 Opportunity Security Tests

Verify:

Authentication Enforcement

Authorization Enforcement

Role Restrictions

Permission Restrictions

Audit Generation

10.22 Opportunity API Tests

Validate:

Create Endpoint

Update Endpoint

Retrieve Endpoint

Search Endpoint

Version Endpoint

History Endpoint

10.23 Opportunity Repository Tests

Verify:

Create Operations

Read Operations

Update Operations

Search Operations

Version Operations

10.24 Opportunity Database Tests

Validate:

Persistence Integrity

Constraint Enforcement

Relationship Integrity

Version Storage

Audit Storage

10.25 Opportunity Workflow Tests

Validate End-To-End Flow:

Create Opportunity



Validate Opportunity



Approve Opportunity



Activate Opportunity



Update Opportunity



Archive Opportunity

10.26 Positive Opportunity Tests

Verify:

Valid Inputs

Valid Workflows

Valid Approvals

Expected Outcomes

10.27 Negative Opportunity Tests

Verify:

Invalid Inputs

Unauthorized Access

Invalid Transitions

Policy Violations

Invalid Relationships

10.28 Boundary Opportunity Tests

Validate:

Maximum Field Lengths

Minimum Field Lengths

Large Descriptions

Large Metadata Sets

Maximum Relationship Counts

10.29 Concurrency Tests

Verify:

Concurrent Updates

Concurrent Reads

Version Conflict Handling

Transaction Integrity

Data Consistency

10.30 Performance Tests

Validate:

Create Latency

Update Latency

Search Latency

Version Retrieval Latency

Workflow Completion Time

10.31 Resilience Tests

Verify:

Database Failure Recovery

Cache Failure Recovery

Transaction Rollbacks

Graceful Error Handling

10.32 Test Data Requirements

Use:

Valid Opportunities

Invalid Opportunities

Archived Opportunities

Large Opportunities

Synthetic Opportunity Data

10.33 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Performance Test Environment

10.34 Coverage Targets

Opportunity Coverage:

100%

Lifecycle Coverage:

100%

Governance Coverage:

100%

Audit Coverage:

100%

10.35 Defect Detection Goals

Detect:

Lifecycle Defects

Validation Defects

Versioning Defects

Search Defects

Governance Defects

Audit Defects

10.36 Pass Criteria

Opportunity Test Passes When:

Lifecycle Executes Correctly

Validation Rules Enforced

Versions Managed Correctly

Governance Rules Enforced

Audit Records Generated

10.37 Failure Criteria

Opportunity Test Fails When:

Invalid Lifecycle Behavior

Data Integrity Failure

Versioning Failure

Governance Failure

Missing Audit Records

10.38 Traceability Requirements

Every Opportunity Test Maps To:

Requirement



Opportunity Component



Workflow



Execution Result

10.39 Auditability Requirements

Maintain:

Execution Reports

Workflow Reports

Coverage Reports

Defect Reports

Audit Reports

10.40 Reproducibility Requirements

Support:

Workflow Replay

Version Replay

Search Replay

Execution Replay

Result Replay

10.41 Governance Rules

TEST-OPP-01

All opportunity workflows require testing.

TEST-OPP-02

Lifecycle validation is mandatory.

TEST-OPP-03

Version validation is mandatory.

TEST-OPP-04

Governance validation is mandatory.

TEST-OPP-05

Audit validation is mandatory.

TEST-OPP-06

Coverage must be measurable.

10.42 Architectural Constraints

AC-TEST-OPP-01

Untested opportunity workflows are prohibited.

AC-TEST-OPP-02

Lifecycle validation cannot be skipped.

AC-TEST-OPP-03

Version integrity must be preserved.

AC-TEST-OPP-04

Coverage gaps require remediation.

AC-TEST-OPP-05

Verification evidence cannot be deleted.

AC-TEST-OPP-06

Governance controls cannot be bypassed.

10.43 Section Summary

This section defines the testing strategy, lifecycle validation requirements, governance controls, version management validation, traceability standards, and audit requirements for the Opportunity Module.

The Opportunity testing layer ensures that opportunity management remains reliable, auditable, governance-compliant, traceable, and consistent throughout the entire lifecycle of an opportunity.

TEST SPEC Progress

🔗 Section 1 — Document Overview & Testing Scope

🔗 Section 2 — Testing Strategy & Methodology

🔗 Section 3 — Test Environment Architecture

🔗 Section 4 — Unit Testing Specifications

🔗 Section 5 — Integration Testing Specifications

☐ Section 6 — API Testing Specifications

☐ Section 7 — Database Testing Specifications

☐ Section 8 — Authentication Testing Specifications

☐ Section 9 — Authorization & RBAC Testing Specifications

☐ Section 10 — Opportunity Module Test Specifications

☐ Section 11 — Authority Module Test Specifications

TEST SPEC V1 — Section 11

Authority Module Test Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

11

Version:

V1

Status:

APPROVED

11.1 Purpose

This section defines the testing specifications for the Authority Module.

The Authority Module is responsible for managing authority sources, authority validation, authority credibility assessment, confidence scoring, governance controls, and auditability.

The objective is to verify:

- Authority lifecycle correctness
- Authority validation
- Confidence score calculation
- Authority governance compliance
- Authority traceability
- Authority auditability

11.2 Testing Objectives

AUTHORITY-TEST-01

Verify authority creation.

AUTHORITY-TEST-02

Verify authority validation.

AUTHORITY-TEST-03

Verify confidence scoring.

AUTHORITY-TEST-04

Verify governance controls.

AUTHORITY-TEST-05

Verify traceability.

AUTHORITY-TEST-06

Verify auditability.

11.3 Scope

Components Covered:

Authority APIs

Authority Services

Authority Repositories

Authority Database Tables

Authority Validators

Confidence Engines

Authority Audit Records

11.4 Authority Lifecycle Validation

Authority Creation

↓

Validation

↓

Verification

↓

Confidence Evaluation



Approval



Operational Usage



Historical Archive

All transitions must be validated.

11.5 Authority Creation Tests

Verify:

Valid Authority Creation

Required Fields

Optional Fields

Metadata Creation

Audit Record Creation

Relationship Creation

11.6 Authority Creation Negative Tests

Verify:

Missing Fields

Invalid Formats

Duplicate Authorities

Malformed Requests

Unauthorized Requests

11.7 Authority Validation Tests

Validate:

Authority Metadata

Authority Source Information

Authority Constraints

Authority Rules

Validation Consistency

11.8 Authority Verification Tests

Verify:

Verification Requests

Verification Results

Verification Updates

Verification Traceability

Verification Audits

11.9 Confidence Score Tests

Validate:

Score Generation

Score Updates

Score Boundaries

Score Persistence

Score Traceability

11.10 Confidence Calculation Accuracy Tests

Verify:

Formula Accuracy

Weight Application

Threshold Evaluation

Rule Evaluation

Result Consistency

11.11 Authority Retrieval Tests

Validate:

Retrieve By ID

Retrieve By Type

Retrieve By Status

Retrieve By Confidence Score

Pagination Support

11.12 Authority Search Tests

Verify:

Keyword Search

Filter Search

Metadata Search

Status Search

Confidence-Based Search

11.13 Authority Status Tests

Validate:

Draft Status

Verified Status

Approved Status

Archived Status

Historical Status

11.14 Authority State Transition Tests

Verify:

Draft → Verified

Verified → Approved

Approved → Archived

Historical Record Creation

Invalid Transition Rejection

11.15 Authority Relationship Tests

Validate:

Authority ↔ Evidence

Authority ↔ Opportunity

Authority ↔ Recommendation

Authority ↔ Audit Records

11.16 Authority Governance Tests

Verify:

Approval Requirements

Governance Restrictions

Policy Enforcement

Compliance Validation

Approval Traceability

11.17 Authority Audit Tests

Validate:

Creation Audits

Validation Audits

Verification Audits

Confidence Audits

Status Change Audits

11.18 Authority Security Tests

Verify:

Authentication Enforcement

Authorization Enforcement

Permission Restrictions

Administrative Controls

Audit Generation

11.19 Authority API Tests

Validate:

Create Authority Endpoint

Update Authority Endpoint

Retrieve Authority Endpoint

Search Authority Endpoint

Verification Endpoint

Confidence Endpoint

11.20 Authority Repository Tests

Verify:

Create Operations

Read Operations

Update Operations

Search Operations

Verification Operations

11.21 Authority Database Tests

Validate:

Persistence Integrity

Constraint Enforcement

Relationship Integrity

Verification Storage

Confidence Storage

Audit Storage

11.22 Authority Workflow Tests

Validate End-To-End Flow:

Create Authority



Validate Authority



Verify Authority



Calculate Confidence



Approve Authority



Operational Usage



Archive Authority

11.23 Positive Authority Tests

Verify:

Valid Inputs

Valid Verification

Valid Confidence Scores

Valid Governance Actions

Expected Outcomes

11.24 Negative Authority Tests

Verify:

Invalid Inputs

Unauthorized Access

Invalid Verification Data

Invalid Confidence Inputs

Policy Violations

11.25 Boundary Authority Tests

Validate:

Maximum Metadata Length

Minimum Metadata Length

Maximum Confidence Values

Minimum Confidence Values

Large Authority Datasets

11.26 Concurrency Tests

Verify:

Concurrent Authority Updates

Concurrent Verification Requests

Concurrent Confidence Calculations

Transaction Integrity

Data Consistency

11.27 Performance Tests

Validate:

Authority Creation Latency

Authority Retrieval Latency

Verification Latency

Confidence Calculation Latency

Search Latency

11.28 Resilience Tests

Verify:

Database Failure Recovery

Cache Failure Recovery

Verification Failure Recovery

Graceful Error Handling

Transaction Rollbacks

11.29 Test Data Requirements

Use:

Valid Authorities

Invalid Authorities

Verified Authorities

Archived Authorities

Synthetic Authority Data

11.30 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Performance Test Environment

11.31 Coverage Targets

Authority Coverage:

100%

Verification Coverage:

100%

Confidence Coverage:

100%

Governance Coverage:

100%

11.32 Defect Detection Goals

Detect:

Validation Defects

Verification Defects

Confidence Defects

Governance Defects

Relationship Defects

Audit Defects

11.33 Pass Criteria

Authority Test Passes When:

Validation Successful

Verification Successful

Confidence Correct

Governance Rules Enforced

Audit Records Generated

11.34 Failure Criteria

Authority Test Fails When:

Invalid Validation Behavior

Incorrect Verification Results

Incorrect Confidence Results

Governance Violations

Missing Audit Records

11.35 Traceability Requirements

Every Authority Test Maps To:

Requirement

↓

Authority Component

↓

Workflow



Execution Result

11.36 Auditability Requirements

Maintain:

Execution Reports

Verification Reports

Confidence Reports

Coverage Reports

Defect Reports

Audit Reports

11.37 Reproducibility Requirements

Support:

Authority Replay

Verification Replay

Confidence Replay

Execution Replay

Result Replay

11.38 Governance Rules

TEST-AUTHORITY-01

All authority workflows require testing.

TEST-AUTHORITY-02

Verification validation is mandatory.

TEST-AUTHORITY-03

Confidence validation is mandatory.

TEST-AUTHORITY-04

Governance validation is mandatory.

TEST-AUTHORITY-05

Audit validation is mandatory.

TEST-AUTHORITY-06

Coverage must be measurable.

11.39 Architectural Constraints

AC-TEST-AUTHORITY-01

Untested authority workflows are prohibited.

AC-TEST-AUTHORITY-02

Verification validation cannot be skipped.

AC-TEST-AUTHORITY-03

Confidence integrity must be preserved.

AC-TEST-AUTHORITY-04

Coverage gaps require remediation.

AC-TEST-AUTHORITY-05

Verification evidence cannot be deleted.

AC-TEST-AUTHORITY-06

Governance controls cannot be bypassed.

11.40 Section Summary

This section defines the testing strategy, validation requirements, confidence scoring verification, governance controls, traceability standards, and audit requirements for the Authority Module.

The Authority testing layer ensures that authority evaluation remains reliable, explainable, auditable, governance-compliant, traceable, and consistent throughout the authority lifecycle.

TEST SPEC Progress

- ❏ Section 1 — Document Overview & Testing Scope
- ❏ Section 2 — Testing Strategy & Methodology
- ❏ Section 3 — Test Environment Architecture
- ❏ Section 4 — Unit Testing Specifications
- ❏ Section 5 — Integration Testing Specifications
- ❏ Section 6 — API Testing Specifications
- ❏ Section 7 — Database Testing Specifications
- ❏ Section 8 — Authentication Testing Specifications
- ❏ Section 9 — Authorization & RBAC Testing Specifications
- ❏ Section 10 — Opportunity Module Test Specifications
- ❏ Section 11 — Authority Module Test Specifications

- ❏ Section 12 — Evidence Module Test Specifications

TEST SPEC V1 — Section 12

Evidence Module Test Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

12

Version:

V1

Status:

APPROVED

12.1 Purpose

This section defines the testing specifications for the Evidence Module.

The Evidence Module is responsible for evidence ingestion, validation, versioning, lineage management, relationship tracking, governance compliance, and auditability.

The objective is to verify:

- Evidence lifecycle correctness
- Evidence validation
- Evidence versioning
- Evidence lineage integrity
- Governance compliance
- Auditability

12.2 Testing Objectives

EVID-TEST-01

Verify evidence creation.

EVID-TEST-02

Verify evidence validation.

EVID-TEST-03

Verify evidence versioning.

EVID-TEST-04

Verify evidence lineage.

EVID-TEST-05

Verify governance controls.

EVID-TEST-06

Verify auditability.

12.3 Scope

Components Covered:

Evidence APIs

Evidence Services

Evidence Repositories

Evidence Database Tables

Evidence Validators

Evidence Lineage Engine

Evidence Audit Records

Evidence Version Records

12.4 Evidence Lifecycle Validation

Evidence Creation

↓

Validation



Approval



Operational Usage



Version Updates



Historical Archive

All lifecycle transitions must be validated.

12.5 Evidence Creation Tests

Verify:

Valid Evidence Creation

Required Fields

Optional Fields

Metadata Creation

Audit Record Creation

Relationship Creation

12.6 Evidence Creation Negative Tests

Verify:

Missing Required Fields

Invalid Formats

Malformed Requests

Duplicate Evidence

Unauthorized Requests

12.7 Evidence Validation Tests

Validate:

Evidence Structure

Evidence Metadata

Evidence Source Information

Evidence Rules

Validation Consistency

12.8 Evidence Quality Tests

Verify:

Required Attributes

Evidence Completeness

Evidence Consistency

Evidence Integrity

Evidence Accuracy Rules

12.9 Evidence Retrieval Tests

Validate:

Retrieve By ID

Retrieve By Type

Retrieve By Status

Retrieve By Metadata

Pagination Support

12.10 Evidence Search Tests

Verify:

Keyword Search

Filter Search

Metadata Search

Status Search

Relationship Search

12.11 Evidence Versioning Tests

Validate:

Version Creation

Version Retrieval

Version Comparison

Version Replay

Version Integrity

12.12 Evidence Version Consistency Tests

Verify:

Historical Accuracy

Version Ordering

Change Tracking

Rollback Integrity

Version Traceability

12.13 Evidence Lineage Tests

Validate:

Lineage Creation

Lineage Retrieval

Lineage Updates

Lineage Traversal

Lineage Integrity

12.14 Evidence Relationship Tests

Verify:

Evidence $\leftrightarrow$ Opportunity

Evidence $\leftrightarrow$ Authority

Evidence $\leftrightarrow$ Recommendation

Evidence $\leftrightarrow$ Audit Records

Relationship Consistency

12.15 Evidence Status Tests

Validate:

Draft Status

Validated Status

Approved Status

Archived Status

Historical Status

12.16 Evidence State Transition Tests

Verify:

Draft → Validated

Validated → Approved

Approved → Archived

Historical Record Creation

Invalid Transition Rejection

12.17 Evidence Governance Tests

Validate:

Approval Requirements

Governance Restrictions

Policy Enforcement

Compliance Validation

Approval Traceability

12.18 Evidence Audit Tests

Verify:

Creation Audits

Validation Audits

Version Audits

Lineage Audits

Status Change Audits

12.19 Evidence Security Tests

Validate:

Authentication Enforcement

Authorization Enforcement

Permission Restrictions

Administrative Controls

Audit Generation

12.20 Evidence API Tests

Verify:

Create Evidence Endpoint

Update Evidence Endpoint

Retrieve Evidence Endpoint

Search Evidence Endpoint

Version Endpoint

Lineage Endpoint

12.21 Evidence Repository Tests

Validate:

Create Operations

Read Operations

Update Operations

Search Operations

Version Operations

Lineage Operations

12.22 Evidence Database Tests

Verify:

Persistence Integrity

Constraint Enforcement

Relationship Integrity

Version Storage

Lineage Storage

Audit Storage

12.23 Evidence Workflow Tests

Validate End-To-End Flow:

Create Evidence



Validate Evidence



Approve Evidence



Link Evidence



Version Evidence



Archive Evidence

12.24 Positive Evidence Tests

Verify:

Valid Inputs

Valid Validation

Valid Relationships

Valid Governance Actions

Expected Outcomes

12.25 Negative Evidence Tests

Verify:

Invalid Inputs

Unauthorized Access

Invalid Relationships

Invalid Versions

Policy Violations

12.26 Boundary Evidence Tests

Validate:

Maximum Metadata Length

Minimum Metadata Length

Large Evidence Records

Large Relationship Sets

Large Lineage Graphs

12.27 Concurrency Tests

Verify:

Concurrent Updates

Concurrent Version Creation

Concurrent Lineage Updates

Transaction Integrity

Data Consistency

12.28 Performance Tests

Validate:

Evidence Creation Latency

Evidence Retrieval Latency

Search Latency

Version Retrieval Latency

Lineage Traversal Latency

12.29 Resilience Tests

Verify:

Database Failure Recovery

Cache Failure Recovery

Lineage Failure Recovery

Graceful Error Handling

Transaction Rollbacks

12.30 Test Data Requirements

Use:

Valid Evidence

Invalid Evidence

Approved Evidence

Archived Evidence

Synthetic Evidence Data

12.31 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Performance Test Environment

12.32 Coverage Targets

Evidence Coverage:

100%

Version Coverage:

100%

Lineage Coverage:

100%

Governance Coverage:

100%

12.33 Defect Detection Goals

Detect:

Validation Defects

Versioning Defects

Lineage Defects

Relationship Defects

Governance Defects

Audit Defects

12.34 Pass Criteria

Evidence Test Passes When:

Validation Successful

Versioning Correct

Lineage Accurate

Governance Rules Enforced

Audit Records Generated

12.35 Failure Criteria

Evidence Test Fails When:

Invalid Validation Behavior

Version Integrity Failure

Lineage Integrity Failure

Governance Violations

Missing Audit Records

12.36 Traceability Requirements

Every Evidence Test Maps To:

Requirement

↓

Evidence Component

↓

Workflow

↓

Execution Result

12.37 Auditability Requirements

Maintain:

Execution Reports

Validation Reports

Version Reports

Lineage Reports

Coverage Reports

Defect Reports

Audit Reports

12.38 Reproducibility Requirements

Support:

Evidence Replay

Version Replay

Lineage Replay

Execution Replay

Result Replay

12.39 Governance Rules

TEST-EVID-01

All evidence workflows require testing.

TEST-EVID-02

Validation testing is mandatory.

TEST-EVID-03

Versioning validation is mandatory.

TEST-EVID-04

Lineage validation is mandatory.

TEST-EVID-05

Audit validation is mandatory.

TEST-EVID-06

Coverage must be measurable.

12.40 Architectural Constraints

AC-TEST-EVID-01

Untested evidence workflows are prohibited.

AC-TEST-EVID-02

Validation cannot be skipped.

AC-TEST-EVID-03

Lineage integrity must be preserved.

AC-TEST-EVID-04

Coverage gaps require remediation.

AC-TEST-EVID-05

Verification evidence cannot be deleted.

AC-TEST-EVID-06

Governance controls cannot be bypassed.

12.41 Section Summary

This section defines the testing strategy, validation requirements, version management verification, lineage integrity controls, governance requirements, traceability standards, and audit controls for the Evidence Module.

The Evidence testing layer ensures that evidence management remains reliable, explainable, traceable, auditable, governance-compliant, and consistent throughout the complete evidence lifecycle.

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TEST SPEC V1 — Section 13

Retrieval Module Test Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

13

Version:

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APPROVED

13.1 Purpose

This section defines the testing specifications for the Retrieval Module.

The Retrieval Module is responsible for identifying, filtering, ranking retrieval candidates, executing keyword search, semantic search, hybrid retrieval, maintaining search history, and supporting replayability.

The objective is to verify:

- Retrieval correctness
- Search accuracy
- Result consistency
- Replayability
- Governance compliance
- Auditability

13.2 Testing Objectives

RETR-TEST-01

Verify retrieval execution.

RETR-TEST-02

Verify search accuracy.

RETR-TEST-03

Verify retrieval consistency.

RETR-TEST-04

Verify replayability.

RETR-TEST-05

Verify governance controls.

RETR-TEST-06

Verify auditability.

13.3 Scope

Components Covered:

Search APIs

Retrieval Services

Retrieval Repositories

Search History Tables

Embedding Storage

Hybrid Retrieval Engine

Replay Engine

Audit Records

13.4 Retrieval Workflow Validation

Search Request

↓

Query Processing

↓

Keyword Retrieval



Semantic Retrieval



Hybrid Fusion



Result Generation



Persistence



Replay Support

All workflow stages must be validated.

13.5 Search Execution Tests

Verify:

Search Creation

Search Completion

Search Persistence

Search Retrieval

Search Replay

13.6 Search Execution Negative Tests

Verify:

Empty Queries

Malformed Queries

Unauthorized Requests

Invalid Filters

Invalid Parameters

13.7 Keyword Retrieval Tests

Validate:

BM25 Execution

Keyword Matching

Result Ordering

Result Filtering

Result Accuracy

13.8 Semantic Retrieval Tests

Verify:

Embedding Usage

Vector Search

Similarity Scoring

Result Relevance

Result Consistency

13.9 Hybrid Retrieval Tests

Validate:

Keyword Results

Semantic Results

Fusion Logic

Ranking Consistency

Final Result Accuracy

13.10 Query Processing Tests

Verify:

Query Parsing

Query Validation

Query Normalization

Query Constraints

Error Handling

13.11 Search Accuracy Tests

Validate:

Expected Results Returned

Incorrect Results Excluded

Relevant Results Prioritized

Ranking Consistency

Filtering Accuracy

13.12 Search Filter Tests

Verify:

Status Filters

Metadata Filters

Authority Filters

Evidence Filters

Combined Filters

13.13 Search History Tests

Validate:

History Creation

History Retrieval

History Integrity

Timeline Consistency

Audit Consistency

13.14 Retrieval Replay Tests

Verify:

Replay Execution

Replay Accuracy

Replay Consistency

Replay Traceability

Replay Audit Records

13.15 Embedding Tests

Validate:

Embedding Creation

Embedding Storage

Embedding Retrieval

Embedding Consistency

Embedding Versioning

13.16 Search Result Tests

Verify:

Result Completeness

Result Consistency

Result Stability

Result Ordering

Result Metadata

13.17 Retrieval Governance Tests

Validate:

Approved Search Execution

Governance Restrictions

Policy Enforcement

Compliance Validation

Approval Traceability

13.18 Retrieval Audit Tests

Verify:

Search Audits

Replay Audits

Filter Audits

Query Audits

History Audits

13.19 Retrieval Security Tests

Validate:

Authentication Enforcement

Authorization Enforcement

Search Restrictions

Administrative Controls

Audit Generation

13.20 Retrieval API Tests

Verify:

Search Endpoint

Keyword Search Endpoint

Semantic Search Endpoint

Hybrid Search Endpoint

Replay Endpoint

History Endpoint

13.21 Retrieval Repository Tests

Validate:

Create Operations

Read Operations

Search Operations

Replay Operations

History Operations

13.22 Retrieval Database Tests

Verify:

Persistence Integrity

Constraint Enforcement

History Storage

Embedding Storage

Audit Storage

13.23 Retrieval Workflow Tests

Validate End-To-End Flow:

Submit Query

↓

Process Query

↓

Execute Retrieval

↓

Generate Results

↓

Persist Search

↓

Replay Search

13.24 Positive Retrieval Tests

Verify:

Valid Queries

Valid Filters

Valid Search Parameters

Valid Governance State

Expected Outcomes

13.25 Negative Retrieval Tests

Verify:

Invalid Queries

Unauthorized Access

Invalid Filters

Malformed Requests

Policy Violations

13.26 Boundary Retrieval Tests

Validate:

Large Queries

Large Result Sets

Large Embedding Sets

Maximum Filters

Complex Search Conditions

13.27 Concurrency Tests

Verify:

Concurrent Searches

Concurrent Replay Requests

Concurrent History Writes

Transaction Integrity

Data Consistency

13.28 Performance Tests

Validate:

Search Latency

Keyword Retrieval Latency

Semantic Retrieval Latency

Hybrid Retrieval Latency

Replay Latency

13.29 Resilience Tests

Verify:

Database Failure Recovery

Embedding Store Failure Recovery

Cache Failure Recovery

Graceful Error Handling

Transaction Rollbacks

13.30 Test Data Requirements

Use:

Valid Queries

Invalid Queries

Large Datasets

Synthetic Embeddings

Synthetic Search Histories

13.31 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Performance Test Environment

13.32 Coverage Targets

Retrieval Coverage:

100%

Search Coverage:

100%

Replay Coverage:

100%

Governance Coverage:

100%

13.33 Defect Detection Goals

Detect:

Search Defects

Retrieval Defects

Fusion Defects

Replay Defects

Governance Defects

Audit Defects

13.34 Pass Criteria

Retrieval Test Passes When:

Search Executes Correctly

Results Accurate

Replay Accurate

Governance Rules Enforced

Audit Records Generated

13.35 Failure Criteria

Retrieval Test Fails When:

Search Incorrect

Results Inaccurate

Replay Inaccurate

Governance Violations

Missing Audit Records

13.36 Traceability Requirements

Every Retrieval Test Maps To:

Requirement

↓

Retrieval Component

↓

Workflow

↓

Execution Result

13.37 Auditability Requirements

Maintain:

Execution Reports

Search Reports

Replay Reports

Coverage Reports

Defect Reports

Audit Reports

13.38 Reproducibility Requirements

Support:

Search Replay

Query Replay

Execution Replay

Result Replay

History Replay

13.39 Governance Rules

TEST-RETR-01

All retrieval workflows require testing.

TEST-RETR-02

Search accuracy validation is mandatory.

TEST-RETR-03

Replay validation is mandatory.

TEST-RETR-04

Governance validation is mandatory.

TEST-RETR-05

Audit validation is mandatory.

TEST-RETR-06

Coverage must be measurable.

13.40 Architectural Constraints

AC-TEST-RETR-01

Untested retrieval workflows are prohibited.

AC-TEST-RETR-02

Search validation cannot be skipped.

AC-TEST-RETR-03

Replay integrity must be preserved.

AC-TEST-RETR-04

Coverage gaps require remediation.

AC-TEST-RETR-05

Verification evidence cannot be deleted.

AC-TEST-RETR-06

Governance controls cannot be bypassed.

13.41 Section Summary

This section defines the testing strategy, search validation requirements, retrieval verification approach, replay controls, governance requirements, traceability standards, and audit controls for the Retrieval Module.

The Retrieval testing layer ensures that search and retrieval operations remain accurate, reproducible, auditable, governance-compliant, traceable, and consistent across all retrieval executions.

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Section 14 — Ranking Module Test Specifications

TEST SPEC V1 — Section 14

Ranking Module Test Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

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Version:

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APPROVED

14.1 Purpose

This section defines the testing specifications for the Ranking Module.

The Ranking Module is responsible for evaluating retrieved opportunities using approved ranking formulas, applying weighted scoring models, generating ranked results, maintaining ranking history, supporting replayability, and ensuring explainability.

The objective is to verify:

- Ranking correctness
- Formula execution accuracy
- Weight application accuracy
- Ranking reproducibility
- Governance compliance
- Auditability

14.2 Testing Objectives

RANK-TEST-01

Verify ranking execution.

RANK-TEST-02

Verify formula correctness.

RANK-TEST-03

Verify scoring accuracy.

RANK-TEST-04

Verify ranking reproducibility.

RANK-TEST-05

Verify governance controls.

RANK-TEST-06

Verify auditability.

14.3 Scope

Components Covered:

Ranking APIs

Ranking Services

Ranking Repositories

Ranking Database Tables

Ranking Formula Engine

Scoring Engine

Ranking Replay Engine

Audit Records

14.4 Ranking Workflow Validation

Retrieved Opportunities

↓

Score Calculation

↓

Formula Execution



Ranking Generation



Explanation Generation



Persistence



Replay Support

All workflow stages must be validated.

14.5 Ranking Execution Tests

Verify:

Ranking Creation

Ranking Completion

Ranking Persistence

Ranking Retrieval

Ranking Replay

14.6 Ranking Execution Negative Tests

Verify:

Empty Candidate Set

Invalid Inputs

Malformed Requests

Unauthorized Requests

Formula Failures

14.7 Formula Validation Tests

Validate:

Formula Selection

Formula Execution

Formula Version Resolution

Formula Constraints

Formula Consistency

14.8 Formula Accuracy Tests

Verify:

Expected Outputs

Calculation Accuracy

Weight Accuracy

Threshold Accuracy

Result Consistency

14.9 Score Calculation Tests

Validate:

Component Scores

Weighted Scores

Aggregate Scores

Final Scores

Boundary Scores

14.10 Weight Application Tests

Verify:

Weight Assignment

Weight Distribution

Weight Changes

Weight Normalization

Weight Consistency

14.11 Candidate Ranking Tests

Validate:

Candidate Ordering

Tie Resolution

Ranking Stability

Ranking Consistency

Result Determinism

14.12 Ranking Retrieval Tests

Verify:

Retrieve By ID

Retrieve By User

Retrieve By Time

Retrieve By Formula

Pagination Support

14.13 Ranking History Tests

Validate:

History Creation

History Retrieval

History Integrity

Timeline Consistency

Audit Consistency

14.14 Formula Versioning Tests

Verify:

Version Creation

Version Retrieval

Version Selection

Version Replay

Version Integrity

14.15 Ranking Replay Tests

Validate:

Replay Execution

Replay Accuracy

Replay Consistency

Replay Traceability

Replay Audit Records

14.16 Explanation Tests

Verify:

Explanation Generation

Explanation Accuracy

Explanation Completeness

Explanation Consistency

Traceability Support

14.17 Ranking Governance Tests

Validate:

Approved Formula Usage

Governance Restrictions

Policy Enforcement

Compliance Validation

Approval Traceability

14.18 Ranking Audit Tests

Verify:

Execution Audits

Formula Audits

Score Audits

Replay Audits

Version Audits

14.19 Ranking Security Tests

Validate:

Authentication Enforcement

Authorization Enforcement

Formula Access Controls

Administrative Controls

Audit Generation

14.20 Ranking API Tests

Verify:

Execute Ranking Endpoint

Retrieve Ranking Endpoint

Replay Endpoint

Formula Endpoint

History Endpoint

14.21 Ranking Repository Tests

Validate:

Create Operations

Read Operations

Search Operations

Replay Operations

Version Operations

14.22 Ranking Database Tests

Verify:

Persistence Integrity

Constraint Enforcement

History Storage

Formula Storage

Audit Storage

14.23 Ranking Workflow Tests

Validate End-To-End Flow:

Retrieve Candidates

↓

Execute Formula

↓

Calculate Scores



Generate Ranking



Generate Explanation



Persist Results



Replay Ranking

14.24 Positive Ranking Tests

Verify:

Valid Inputs

Valid Formula

Valid Weights

Valid Governance State

Expected Outcomes

14.25 Negative Ranking Tests

Verify:

Invalid Inputs

Unauthorized Access

Invalid Formula

Invalid Weights

Policy Violations

14.26 Boundary Ranking Tests

Validate:

Large Candidate Sets

Maximum Scores

Minimum Scores

Extreme Weight Values

Large Ranking Histories

14.27 Concurrency Tests

Verify:

Concurrent Ranking Requests

Concurrent Formula Updates

Concurrent Replay Requests

Transaction Integrity

Data Consistency

14.28 Performance Tests

Validate:

Ranking Latency

Formula Execution Latency

Replay Latency

History Retrieval Latency

Explanation Generation Latency

14.29 Resilience Tests

Verify:

Database Failure Recovery

Formula Engine Failure Recovery

Cache Failure Recovery

Graceful Error Handling

Transaction Rollbacks

14.30 Test Data Requirements

Use:

Valid Candidates

Invalid Candidates

Multiple Formula Versions

Large Candidate Sets

Synthetic Ranking Data

14.31 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Performance Test Environment

14.32 Coverage Targets

Ranking Coverage:

100%

Formula Coverage:

100%

Replay Coverage:

100%

Governance Coverage:

100%

14.33 Defect Detection Goals

Detect:

Formula Defects

Scoring Defects

Weighting Defects

Replay Defects

Governance Defects

Audit Defects

14.34 Pass Criteria

Ranking Test Passes When:

Formula Executes Correctly

Scores Calculated Correctly

Ranking Order Correct

Replay Accurate

Audit Records Generated

14.35 Failure Criteria

Ranking Test Fails When:

Formula Incorrect

Score Incorrect

Ranking Incorrect

Replay Inaccurate

Missing Audit Records

14.36 Traceability Requirements

Every Ranking Test Maps To:

Requirement



Ranking Component



Workflow



Execution Result

14.37 Auditability Requirements

Maintain:

Execution Reports

Formula Reports

Replay Reports

Coverage Reports

Defect Reports

Audit Reports

14.38 Reproducibility Requirements

Support:

Ranking Replay

Formula Replay

Score Replay

Execution Replay

Result Replay

14.39 Governance Rules

TEST-RANK-01

All ranking workflows require testing.

TEST-RANK-02

Formula validation is mandatory.

TEST-RANK-03

Replay validation is mandatory.

TEST-RANK-04

Governance validation is mandatory.

TEST-RANK-05

Audit validation is mandatory.

TEST-RANK-06

Coverage must be measurable.

14.40 Architectural Constraints

AC-TEST-RANK-01

Untested ranking workflows are prohibited.

AC-TEST-RANK-02

Formula validation cannot be skipped.

AC-TEST-RANK-03

Replay integrity must be preserved.

AC-TEST-RANK-04

Coverage gaps require remediation.

AC-TEST-RANK-05

Verification evidence cannot be deleted.

AC-TEST-RANK-06

Governance controls cannot be bypassed.

14.41 Section Summary

This section defines the testing strategy, formula validation requirements, scoring verification approach, replay controls, governance requirements, traceability standards, and audit controls for the Ranking Module.

The Ranking testing layer ensures that ranking decisions remain accurate, explainable, reproducible, auditable, governance-compliant, and consistent across all ranking executions.

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☐ Section 13 — Retrieval Module Test Specifications

☐ Section 14 — Ranking Module Test Specifications

☐ Section 15 — Recommendation Module Test Specifications

TEST SPEC V1 — Section 15

Recommendation Module Test Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

15

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APPROVED

15.1 Purpose

This section defines the testing specifications for the Recommendation Module.

The Recommendation Module is responsible for generating recommendations from ranked opportunities, producing explainable outcomes, maintaining recommendation traceability, supporting replayability, enforcing governance controls, and preserving auditability.

The objective is to verify:

- Recommendation correctness
- Recommendation explainability
- Recommendation traceability
- Recommendation reproducibility
- Governance compliance
- Auditability

15.2 Testing Objectives

REC-TEST-01

Verify recommendation generation.

REC-TEST-02

Verify explanation generation.

REC-TEST-03

Verify traceability generation.

REC-TEST-04

Verify replayability.

REC-TEST-05

Verify governance controls.

REC-TEST-06

Verify auditability.

15.3 Scope

Components Covered:

Recommendation APIs

Recommendation Services

Recommendation Repositories

Recommendation Database Tables

Explanation Engine

Traceability Engine

Replay Engine

Audit Records

15.4 Recommendation Workflow Validation

Ranked Opportunities

↓

Recommendation Evaluation

↓

Recommendation Selection

↓

Explanation Generation



Traceability Generation



Persistence



Replay Support

All workflow stages must be validated.

15.5 Recommendation Generation Tests

Verify:

Recommendation Creation

Recommendation Persistence

Recommendation Retrieval

Recommendation Replay

Recommendation Consistency

15.6 Recommendation Generation Negative Tests

Verify:

Empty Ranking Inputs

Invalid Inputs

Malformed Requests

Unauthorized Requests

Recommendation Failures

15.7 Recommendation Selection Tests

Validate:

Best Candidate Selection

Selection Consistency

Selection Stability

Tie Resolution

Policy Compliance

15.8 Recommendation Accuracy Tests

Verify:

Expected Recommendation Returned

Incorrect Recommendation Rejected

Consistency Across Replays

Deterministic Behavior

Result Integrity

15.9 Explanation Generation Tests

Validate:

Explanation Creation

Explanation Completeness

Explanation Accuracy

Explanation Consistency

Explanation Persistence

15.10 Explanation Quality Tests

Verify:

Decision Factors Included

Ranking Factors Included

Authority Factors Included

Evidence Factors Included

Governance Factors Included

15.11 Recommendation Traceability Tests

Validate:

Trace Creation

Trace Retrieval

Trace Completeness

Trace Consistency

Trace Persistence

15.12 Recommendation Lineage Tests

Verify:

Recommendation Source Tracking

Ranking Source Tracking

Evidence Source Tracking

Authority Source Tracking

Decision Lineage Integrity

15.13 Recommendation Retrieval Tests

Validate:

Retrieve By ID

Retrieve By User

Retrieve By Time

Retrieve By Status

Pagination Support

15.14 Recommendation History Tests

Verify:

History Creation

History Retrieval

History Integrity

Timeline Consistency

Audit Consistency

15.15 Recommendation Replay Tests

Validate:

Replay Execution

Replay Accuracy

Replay Consistency

Replay Traceability

Replay Audit Records

15.16 Recommendation Governance Tests

Verify:

Approved Recommendation Usage

Governance Restrictions

Policy Enforcement

Compliance Validation

Approval Traceability

15.17 Recommendation Audit Tests

Validate:

Recommendation Audits

Explanation Audits

Replay Audits

Traceability Audits

History Audits

15.18 Recommendation Security Tests

Verify:

Authentication Enforcement

Authorization Enforcement

Recommendation Restrictions

Administrative Controls

Audit Generation

15.19 Recommendation API Tests

Validate:

Generate Recommendation Endpoint

Retrieve Recommendation Endpoint

Explanation Endpoint

Trace Endpoint

Replay Endpoint

History Endpoint

15.20 Recommendation Repository Tests

Verify:

Create Operations

Read Operations

Search Operations

Replay Operations

History Operations

15.21 Recommendation Database Tests

Validate:

Persistence Integrity

Constraint Enforcement

History Storage

Trace Storage

Audit Storage

15.22 Recommendation Workflow Tests

Validate End-To-End Flow:

Retrieve Ranked Results



Generate Recommendation



Generate Explanation



Generate Traceability



Persist Recommendation



Replay Recommendation

15.23 Positive Recommendation Tests

Verify:

Valid Inputs

Valid Ranking Results

Valid Governance State

Valid Traceability

Expected Outcomes

15.24 Negative Recommendation Tests

Verify:

Invalid Inputs

Unauthorized Access

Missing Ranking Results

Traceability Failures

Policy Violations

15.25 Boundary Recommendation Tests

Validate:

Large Candidate Sets

Large Explanations

Large Trace Graphs

Maximum Recommendation History

Complex Decision Trees

15.26 Concurrency Tests

Verify:

Concurrent Recommendation Requests

Concurrent Replay Requests

Concurrent History Writes

Transaction Integrity

Data Consistency

15.27 Performance Tests

Validate:

Recommendation Latency

Explanation Generation Latency

Trace Generation Latency

Replay Latency

History Retrieval Latency

15.28 Resilience Tests

Verify:

Database Failure Recovery

Explanation Engine Failure Recovery

Traceability Engine Failure Recovery

Graceful Error Handling

Transaction Rollbacks

15.29 Test Data Requirements

Use:

Valid Recommendations

Invalid Recommendations

Large Ranking Sets

Synthetic Explanations

Synthetic Traceability Data

15.30 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Performance Test Environment

15.31 Coverage Targets

Recommendation Coverage:

100%

Explanation Coverage:

100%

Traceability Coverage:

100%

Governance Coverage:

100%

15.32 Defect Detection Goals

Detect:

Recommendation Defects

Explanation Defects

Traceability Defects

Replay Defects

Governance Defects

Audit Defects

15.33 Pass Criteria

Recommendation Test Passes When:

Recommendation Correct

Explanation Accurate

Traceability Complete

Replay Accurate

Audit Records Generated

15.34 Failure Criteria

Recommendation Test Fails When:

Recommendation Incorrect

Explanation Inaccurate

Traceability Incomplete

Replay Inaccurate

Missing Audit Records

15.35 Traceability Requirements

Every Recommendation Test Maps To:

Requirement

↓

Recommendation Component

↓

Workflow

↓

Execution Result

15.36 Auditability Requirements

Maintain:

Execution Reports

Recommendation Reports

Explanation Reports

Replay Reports

Coverage Reports

Defect Reports

Audit Reports

15.37 Reproducibility Requirements

Support:

Recommendation Replay

Explanation Replay

Trace Replay

Execution Replay

Result Replay

15.38 Governance Rules

TEST-REC-01

All recommendation workflows require testing.

TEST-REC-02

Explanation validation is mandatory.

TEST-REC-03

Traceability validation is mandatory.

TEST-REC-04

Governance validation is mandatory.

TEST-REC-05

Audit validation is mandatory.

TEST-REC-06

Coverage must be measurable.

15.39 Architectural Constraints

AC-TEST-REC-01

Untested recommendation workflows are prohibited.

AC-TEST-REC-02

Explanation validation cannot be skipped.

AC-TEST-REC-03

Traceability integrity must be preserved.

AC-TEST-REC-04

Coverage gaps require remediation.

AC-TEST-REC-05

Verification evidence cannot be deleted.

AC-TEST-REC-06

Governance controls cannot be bypassed.

15.40 Section Summary

This section defines the testing strategy, recommendation validation requirements, explanation verification approach, traceability controls, governance requirements, audit standards, and replay requirements for the Recommendation Module.

The Recommendation testing layer ensures that recommendation outcomes remain accurate, explainable, reproducible, traceable, auditable, governance-compliant, and consistent across all recommendation executions.

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- ❏ Section 15 — Recommendation Module Test Specifications

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TEST SPEC V1 — Section 16

Governance Module Test Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

16

Version:

V1

Status:

APPROVED

16.1 Purpose

This section defines the testing specifications for the Governance Module.

The Governance Module is responsible for policy enforcement, approval workflows, compliance validation, governance decision tracking, exception handling, and organizational control mechanisms across TODP.

The objective is to verify:

- Governance policy enforcement
- Approval workflow correctness
- Compliance validation
- Governance traceability
- Governance auditability
- Governance reproducibility

16.2 Testing Objectives

GOV-TEST-01

Verify policy enforcement.

GOV-TEST-02

Verify approval workflows.

GOV-TEST-03

Verify compliance validation.

GOV-TEST-04

Verify governance traceability.

GOV-TEST-05

Verify governance auditability.

GOV-TEST-06

Verify governance reproducibility.

16.3 Scope

Components Covered:

Governance APIs

Governance Services

Governance Repositories

Governance Database Tables

Policy Engine

Approval Engine

Compliance Engine

Governance Audit Records

16.4 Governance Workflow Validation

Policy Definition

↓

Policy Evaluation

↓

Approval Decision

↓

Compliance Validation

↓

Governance Action

↓

Audit Recording

↓

Replay Support

All workflow stages must be validated.

16.5 Policy Management Tests

Verify:

Policy Creation

Policy Updates

Policy Retrieval

Policy Versioning

Policy Deactivation

16.6 Policy Validation Tests

Validate:

Policy Syntax

Policy Semantics

Policy Constraints

Policy Dependencies

Policy Consistency

16.7 Policy Enforcement Tests

Verify:

Allow Decisions

Deny Decisions

Conditional Decisions

Exception Handling

Rule Prioritization

16.8 Policy Conflict Tests

Validate:

Conflict Detection

Conflict Resolution

Priority Evaluation

Override Handling

Decision Consistency

16.9 Approval Workflow Tests

Verify:

Approval Request Creation

Approval Assignment

Approval Decisions

Approval Completion

Approval Rejection

16.10 Multi-Level Approval Tests

Validate:

Single-Level Approval

Multi-Level Approval

Sequential Approval

Parallel Approval

Escalation Handling

16.11 Compliance Validation Tests

Verify:

Compliance Rule Execution

Compliance Result Generation

Compliance Violations

Compliance Reporting

Compliance Traceability

16.12 Compliance Rule Tests

Validate:

Mandatory Rules

Conditional Rules

Threshold Rules

Policy-Based Rules

Cross-Module Rules

16.13 Governance Decision Tests

Verify:

Decision Creation

Decision Retrieval

Decision History

Decision Integrity

Decision Consistency

16.14 Governance Exception Tests

Validate:

Exception Creation

Exception Approval

Exception Expiration

Exception Revocation

Exception Auditability

16.15 Governance Versioning Tests

Verify:

Policy Version Creation

Policy Version Retrieval

Policy Version Replay

Policy Version Integrity

Version Traceability

16.16 Governance Replay Tests

Validate:

Replay Execution

Replay Accuracy

Replay Consistency

Replay Traceability

Replay Audit Records

16.17 Governance Traceability Tests

Verify:

Requirement Traceability

Decision Traceability

Approval Traceability

Policy Traceability

Compliance Traceability

16.18 Governance Audit Tests

Validate:

Policy Audits

Approval Audits

Compliance Audits

Decision Audits

Replay Audits

16.19 Governance Security Tests

Verify:

Authentication Enforcement

Authorization Enforcement

Administrative Restrictions

Policy Access Controls

Audit Generation

16.20 Governance API Tests

Validate:

Policy Endpoint

Approval Endpoint

Compliance Endpoint

Decision Endpoint

Replay Endpoint

Audit Endpoint

16.21 Governance Repository Tests

Verify:

Create Operations

Read Operations

Update Operations

Search Operations

Replay Operations

16.22 Governance Database Tests

Validate:

Persistence Integrity

Constraint Enforcement

Policy Storage

Approval Storage

Audit Storage

16.23 Governance Workflow Tests

Validate End-To-End Flow:

Create Policy



Evaluate Policy



Generate Approval Request



Approve/Reject Request



Validate Compliance



Record Governance Decision



Replay Governance Flow

16.24 Positive Governance Tests

Verify:

Valid Policies

Valid Approvals

Valid Compliance States

Valid Governance Decisions

Expected Outcomes

16.25 Negative Governance Tests

Verify:

Invalid Policies

Unauthorized Access

Invalid Approvals

Compliance Violations

Policy Violations

16.26 Boundary Governance Tests

Validate:

Large Policy Sets

Large Approval Chains

Large Compliance Reports

Maximum Rule Counts

Complex Governance Scenarios

16.27 Concurrency Tests

Verify:

Concurrent Policy Updates

Concurrent Approval Requests

Concurrent Compliance Evaluations

Transaction Integrity

Data Consistency

16.28 Performance Tests

Validate:

Policy Evaluation Latency

Approval Workflow Latency

Compliance Validation Latency

Replay Latency

Decision Retrieval Latency

16.29 Resilience Tests

Verify:

Database Failure Recovery

Policy Engine Failure Recovery

Approval Engine Failure Recovery

Graceful Error Handling

Transaction Rollbacks

16.30 Test Data Requirements

Use:

Valid Policies

Invalid Policies

Approval Scenarios

Compliance Scenarios

Synthetic Governance Data

16.31 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Performance Test Environment

Governance Validation Environment

16.32 Coverage Targets

Governance Coverage:

100%

Policy Coverage:

100%

Approval Coverage:

100%

Compliance Coverage:

100%

16.33 Defect Detection Goals

Detect:

Policy Defects

Approval Defects

Compliance Defects

Replay Defects

Governance Defects

Audit Defects

16.34 Pass Criteria

Governance Test Passes When:

Policies Enforced Correctly

Approvals Executed Correctly

Compliance Validated Correctly

Replay Accurate

Audit Records Generated

16.35 Failure Criteria

Governance Test Fails When:

Policy Enforcement Incorrect

Approval Workflow Incorrect

Compliance Validation Incorrect

Replay Inaccurate

Missing Audit Records

16.36 Traceability Requirements

Every Governance Test Maps To:

Requirement

↓

Governance Component

↓

Workflow



Execution Result

16.37 Auditability Requirements

Maintain:

Execution Reports

Policy Reports

Approval Reports

Compliance Reports

Coverage Reports

Defect Reports

Audit Reports

16.38 Reproducibility Requirements

Support:

Policy Replay

Approval Replay

Compliance Replay

Execution Replay

Result Replay

16.39 Governance Rules

TEST-GOV-01

All governance workflows require testing.

TEST-GOV-02

Policy validation is mandatory.

TEST-GOV-03

Approval validation is mandatory.

TEST-GOV-04

Compliance validation is mandatory.

TEST-GOV-05

Audit validation is mandatory.

TEST-GOV-06

Coverage must be measurable.

16.40 Architectural Constraints

AC-TEST-GOV-01

Untested governance workflows are prohibited.

AC-TEST-GOV-02

Policy validation cannot be skipped.

AC-TEST-GOV-03

Compliance integrity must be preserved.

AC-TEST-GOV-04

Coverage gaps require remediation.

AC-TEST-GOV-05

Verification evidence cannot be deleted.

AC-TEST-GOV-06

Governance controls cannot be bypassed.

16.41 Section Summary

This section defines the testing strategy, policy validation requirements, approval workflow verification approach, compliance controls, governance traceability standards, audit requirements, and replay controls for the Governance Module.

The Governance testing layer ensures that governance decisions remain consistent, enforceable, auditable, reproducible, traceable, and compliant with organizational policies and platform controls.

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- 🔗 Section 5 — Integration Testing Specifications
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- 🔗 Section 16 — Governance Module Test Specifications

- 🔗 Section 17 — Audit & Replay Testing Specifications

TEST SPEC V1 — Section 17

Audit & Replay Testing Specifications

Document:

Test Specification (TEST SPEC V1)

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17.1 Purpose

This section defines the testing specifications for Audit and Replay capabilities within TODP.

The Audit & Replay subsystem is responsible for preserving decision history, maintaining traceability, supporting governance reviews, enabling forensic analysis, and reproducing historical system behavior.

The objective is to verify:

- Audit record integrity
- Replay accuracy
- Traceability completeness
- Historical reproducibility

- Governance compliance
- Forensic reliability

17.2 Testing Objectives

AUDIT-TEST-01

Verify audit record creation.

AUDIT-TEST-02

Verify audit record integrity.

AUDIT-TEST-03

Verify replay accuracy.

AUDIT-TEST-04

Verify traceability completeness.

AUDIT-TEST-05

Verify governance compliance.

AUDIT-TEST-06

Verify forensic reproducibility.

17.3 Scope

Components Covered:

Audit APIs

Audit Services

Audit Repositories

Audit Database Tables

Snapshot Engine

Replay Engine

Correlation Engine

Governance Audit Records

17.4 Audit Workflow Validation

Business Event



Audit Generation



Snapshot Creation



Correlation Assignment



Persistence



Retrieval



Replay

All workflow stages must be validated.

17.5 Audit Record Creation Tests

Verify:

Audit Creation

Metadata Creation

Timestamp Recording

Actor Recording

Event Recording

17.6 Audit Record Integrity Tests

Validate:

Record Completeness

Field Accuracy

Data Consistency

Immutable Storage

Integrity Preservation

17.7 Audit Retrieval Tests

Verify:

Retrieve By ID

Retrieve By User

Retrieve By Event

Retrieve By Time Range

Pagination Support

17.8 Audit Search Tests

Validate:

Keyword Search

Metadata Search

Correlation Search

Workflow Search

Advanced Filtering

17.9 Snapshot Creation Tests

Verify:

Snapshot Generation

Snapshot Persistence

Snapshot Retrieval

Snapshot Integrity

Snapshot Traceability

17.10 Snapshot Accuracy Tests

Validate:

Data Accuracy

State Accuracy

Version Accuracy

Relationship Accuracy

Replay Readiness

17.11 Correlation Tests

Verify:

Correlation ID Creation

Correlation Persistence

Cross-Module Correlation

Workflow Correlation

Traceability Integrity

17.12 Audit Timeline Tests

Validate:

Timeline Creation

Timeline Retrieval

Chronological Ordering

Event Integrity

History Consistency

17.13 Replay Execution Tests

Verify:

Replay Initiation

Replay Completion

Replay Accuracy

Replay Persistence

Replay Audit Records

17.14 Replay Accuracy Tests

Validate:

Result Reproduction

State Reproduction

Decision Reproduction

Formula Reproduction

Trace Reproduction

17.15 Replay Consistency Tests

Verify:

Repeatable Results

Deterministic Outputs

Historical Consistency

Version Consistency

Data Consistency

17.16 Replay Versioning Tests

Validate:

Version Resolution

Version Retrieval

Version Replay

Historical Accuracy

Version Integrity

17.17 Replay Traceability Tests

Verify:

Input Traceability

Decision Traceability

Evidence Traceability

Authority Traceability

Output Traceability

17.18 Governance Audit Tests

Validate:

Governance Decision Audits

Approval Audits

Compliance Audits

Policy Audits

Replay Audits

17.19 Audit Security Tests

Verify:

Authentication Enforcement

Authorization Enforcement

Audit Access Restrictions

Administrative Controls

Audit Logging

17.20 Replay Security Tests

Validate:

Authorized Replay Access

Restricted Replay Access

Replay Audit Generation

Replay Access Controls

17.21 Audit API Tests

Verify:

Audit Retrieval Endpoint

Audit Search Endpoint

Snapshot Endpoint

Correlation Endpoint

History Endpoint

17.22 Replay API Tests

Validate:

Replay Execution Endpoint

Replay Status Endpoint

Replay History Endpoint

Replay Retrieval Endpoint

17.23 Audit Repository Tests

Verify:

Create Operations

Read Operations

Search Operations

Correlation Operations

History Operations

17.24 Replay Repository Tests

Validate:

Replay Storage

Replay Retrieval

Replay Search

Replay History

Replay Persistence

17.25 Audit Database Tests

Verify:

Persistence Integrity

Constraint Enforcement

Snapshot Storage

Correlation Storage

Audit Storage

17.26 Replay Workflow Tests

Validate End-To-End Flow:

Historical Request

↓

Audit Retrieval

↓

Snapshot Resolution



Version Resolution



Replay Execution



Result Validation



Audit Generation

17.27 Positive Audit Tests

Verify:

Valid Events

Valid Snapshots

Valid Correlations

Expected Outcomes

Audit Integrity

17.28 Negative Audit Tests

Verify:

Missing Records

Corrupt Records

Unauthorized Access

Invalid Queries

Policy Violations

17.29 Positive Replay Tests

Validate:

Valid Replay Requests

Valid Versions

Valid Historical Data

Expected Outcomes

Replay Accuracy

17.30 Negative Replay Tests

Verify:

Missing Versions

Corrupt Snapshots

Unauthorized Replay

Invalid Replay Requests

Governance Violations

17.31 Boundary Tests

Validate:

Large Audit Histories

Large Snapshot Volumes

Large Replay Requests

Complex Correlation Graphs

Extended Timelines

17.32 Concurrency Tests

Verify:

Concurrent Audit Writes

Concurrent Replay Requests

Concurrent Snapshot Creation

Transaction Integrity

Data Consistency

17.33 Performance Tests

Validate:

Audit Write Latency

Audit Retrieval Latency

Replay Latency

Correlation Retrieval Latency

Snapshot Retrieval Latency

17.34 Resilience Tests

Verify:

Database Failure Recovery

Replay Engine Failure Recovery

Snapshot Failure Recovery

Graceful Error Handling

Transaction Rollbacks

17.35 Test Data Requirements

Use:

Valid Audit Records

Invalid Audit Records

Historical Snapshots

Multiple Versions

Synthetic Replay Scenarios

17.36 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Governance Validation Environment

Performance Test Environment

17.37 Coverage Targets

Audit Coverage:

100%

Replay Coverage:

100%

Traceability Coverage:

100%

Governance Coverage:

100%

17.38 Defect Detection Goals

Detect:

Audit Defects

Replay Defects

Traceability Defects

Snapshot Defects

Governance Defects

Integrity Defects

17.39 Pass Criteria

Audit & Replay Test Passes When:

Audit Records Accurate

Snapshots Accurate

Replay Accurate

Traceability Complete

Audit Records Generated

17.40 Failure Criteria

Audit & Replay Test Fails When:

Audit Records Incorrect

Replay Inaccurate

Traceability Incomplete

Snapshot Corrupted

Missing Audit Records

17.41 Traceability Requirements

Every Audit & Replay Test Maps To:

Requirement



Audit Component



Replay Component



Execution Result

17.42 Auditability Requirements

Maintain:

Execution Reports

Audit Reports

Replay Reports

Traceability Reports

Coverage Reports

Defect Reports

17.43 Reproducibility Requirements

Support:

Audit Replay

Snapshot Replay

Version Replay

Execution Replay

Result Replay

17.44 Governance Rules

TEST-AUDIT-01

All audit workflows require testing.

TEST-AUDIT-02

Replay validation is mandatory.

TEST-AUDIT-03

Traceability validation is mandatory.

TEST-AUDIT-04

Snapshot integrity validation is mandatory.

TEST-AUDIT-05

Audit evidence must be retained.

TEST-AUDIT-06

Coverage must be measurable.

17.45 Architectural Constraints

AC-TEST-AUDIT-01

Untested audit workflows are prohibited.

AC-TEST-AUDIT-02

Replay validation cannot be skipped.

AC-TEST-AUDIT-03

Audit integrity must be preserved.

AC-TEST-AUDIT-04

Coverage gaps require remediation.

AC-TEST-AUDIT-05

Verification evidence cannot be deleted.

AC-TEST-AUDIT-06

Governance controls cannot be bypassed.

17.46 Section Summary

This section defines the testing strategy, audit validation requirements, replay verification approach, traceability controls, forensic requirements, governance standards, and reproducibility controls for the Audit & Replay subsystem.

The Audit & Replay testing layer ensures that historical decisions remain reproducible, traceable, auditable, governance-compliant, and forensically reliable throughout the lifecycle of TODP.

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TEST SPEC V1 — Section 18

Security Testing Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

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APPROVED

18.1 Purpose

This section defines the security testing specifications for TODP.

Security testing validates that the platform protects data, users, services, infrastructure, and governance processes from unauthorized access, misuse, corruption, disclosure, and operational compromise.

The objective is to verify:

- Confidentiality
- Integrity
- Availability
- Access control enforcement
- Security resilience
- Governance compliance

18.2 Security Testing Objectives

SEC-TEST-01

Verify authentication security.

SEC-TEST-02

Verify authorization security.

SEC-TEST-03

Verify data protection controls.

SEC-TEST-04

Verify infrastructure security.

SEC-TEST-05

Verify governance security.

SEC-TEST-06

Verify security auditability.

18.3 Scope

Components Covered:

Authentication Systems

Authorization Systems

APIs

Databases

Redis

Infrastructure

Secrets Management

Audit Systems

Governance Systems

CI/CD Pipelines

18.4 Security Validation Model

Threat



Control



Validation



Detection



Response



Audit

All security controls must be validated.

18.5 Authentication Security Tests

Verify:

Credential Validation

Password Security

Session Security

Token Security

Account Protection

18.6 Authorization Security Tests

Validate:

RBAC Enforcement

Permission Validation

Access Restrictions

Administrative Controls

Privilege Separation

18.7 Password Security Tests

Verify:

Password Policies

Password Complexity

Password Reuse Restrictions

Hashing Controls

Credential Protection

18.8 Session Security Tests

Validate:

Session Creation

Session Expiration

Session Revocation

Session Isolation

Session Protection

18.9 JWT Security Tests

Verify:

Token Integrity

Token Signature Validation

Token Expiration

Claim Validation

Replay Prevention

18.10 Input Validation Tests

Validate:

Input Sanitization

Boundary Validation

Format Validation

Malicious Input Rejection

Validation Consistency

18.11 API Security Tests

Verify:

Authentication Enforcement

Authorization Enforcement

Request Validation

Response Security

Rate Limiting

18.12 Database Security Tests

Validate:

Database Permissions

Access Restrictions

Encryption Support

Audit Logging

Connection Security

18.13 Redis Security Tests

Verify:

Access Controls

Connection Security

Data Protection

Key Isolation

Credential Protection

18.14 Secrets Management Tests

Validate:

Secret Storage

Secret Access Controls

Secret Rotation

Secret Retrieval

Secret Auditability

18.15 Encryption Tests

Verify:

Encryption At Rest

Encryption In Transit

Key Usage

Key Management

Encryption Integrity

18.16 Audit Security Tests

Validate:

Audit Protection

Audit Integrity

Audit Access Controls

Audit Retention

Audit Traceability

18.17 Governance Security Tests

Verify:

Governance Access Controls

Policy Protection

Approval Protection

Compliance Protection

Administrative Restrictions

18.18 Privilege Escalation Tests

Validate:

Role Escalation Prevention

Permission Escalation Prevention

Administrative Boundary Protection

Governance Protection

18.19 Unauthorized Access Tests

Verify:

Resource Protection

Endpoint Protection

Administrative Protection

Data Protection

Replay Protection

18.20 Account Enumeration Tests

Validate:

User Enumeration Protection

Error Message Controls

Identity Disclosure Prevention

Response Consistency

18.21 Session Hijacking Tests

Verify:

Session Protection

Token Protection

Session Validation

Revocation Enforcement

18.22 Injection Resistance Tests

Validate:

SQL Injection Protection

Command Injection Protection

Template Injection Protection

Input Manipulation Resistance

18.23 Cross-Site Attack Tests

Verify:

Cross-Site Request Protection

Cross-Origin Controls

Request Validation

Security Header Enforcement

18.24 Security Header Tests

Validate:

Transport Security Headers

Content Security Policies

Frame Restrictions

Origin Controls

Browser Protections

18.25 Data Exposure Tests

Verify:

Sensitive Data Protection

Error Message Protection

Metadata Protection

Credential Protection

Secret Protection

18.26 Logging Security Tests

Validate:

Security Event Logging

Authentication Logging

Authorization Logging

Governance Logging

Audit Logging

18.27 Security Monitoring Tests

Verify:

Alert Generation

Security Event Detection

Threshold Detection

Anomaly Detection

Response Triggers

18.28 Vulnerability Assessment Tests

Validate:

Dependency Reviews

Configuration Reviews

Infrastructure Reviews

Application Reviews

Security Baseline Validation

18.29 Penetration Testing Requirements

Verify:

Authentication Controls

Authorization Controls

API Controls

Governance Controls

Audit Controls

18.30 Security Configuration Tests

Validate:

Environment Configuration

Network Configuration

Database Configuration

Service Configuration

Infrastructure Configuration

18.31 Backup Security Tests

Verify:

Backup Encryption

Backup Access Controls

Backup Integrity

Backup Retention

Recovery Security

18.32 Incident Response Tests

Validate:

Detection Procedures

Notification Procedures

Escalation Procedures

Response Procedures

Audit Procedures

18.33 Security Workflow Tests

Validate End-To-End Flow:

Authentication



Authorization



Access Request



Security Validation



Audit Generation



Monitoring



Incident Response

18.34 Positive Security Tests

Verify:

Valid Access

Valid Tokens

Valid Permissions

Valid Security Configurations

Expected Outcomes

18.35 Negative Security Tests

Verify:

Unauthorized Access

Privilege Escalation Attempts

Invalid Tokens

Malformed Requests

Policy Violations

18.36 Boundary Security Tests

Validate:

Large Request Volumes

Large Token Sets

Large Audit Volumes

Complex Access Models

Extreme Workloads

18.37 Concurrency Tests

Verify:

Concurrent Logins

Concurrent Access Requests

Concurrent Permission Evaluations

Concurrent Security Events

Data Consistency

18.38 Performance Tests

Validate:

Authentication Latency

Authorization Latency

Encryption Latency

Audit Latency

Monitoring Latency

18.39 Resilience Tests

Verify:

Security Service Failures

Database Failures

Cache Failures

Monitoring Failures

Graceful Recovery

18.40 Test Data Requirements

Use:

Valid Security Data

Invalid Security Data

Expired Credentials

Revoked Tokens

Synthetic Attack Scenarios

18.41 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Security Test Environment

Governance Validation Environment

18.42 Coverage Targets

Security Coverage:

100%

Authentication Coverage:

100%

Authorization Coverage:

100%

Governance Coverage:

100%

18.43 Defect Detection Goals

Detect:

Authentication Defects

Authorization Defects

Encryption Defects

Configuration Defects

Governance Defects

Audit Defects

18.44 Pass Criteria

Security Test Passes When:

Controls Enforced Correctly

Unauthorized Access Prevented

Data Protected

Monitoring Operational

Audit Records Generated

18.45 Failure Criteria

Security Test Fails When:

Unauthorized Access Possible

Control Failure Detected

Sensitive Data Exposed

Governance Controls Bypassed

Audit Records Missing

18.46 Traceability Requirements

Every Security Test Maps To:

Requirement



Security Control



Execution Result



Verification Record

18.47 Auditability Requirements

Maintain:

Security Reports

Execution Reports

Coverage Reports

Defect Reports

Audit Reports

Incident Reports

18.48 Reproducibility Requirements

Support:

Security Replay

Incident Replay

Execution Replay

Result Replay

Validation Replay

18.49 Governance Rules

TEST-SEC-01

All security controls require testing.

TEST-SEC-02

Access control validation is mandatory.

TEST-SEC-03

Data protection validation is mandatory.

TEST-SEC-04

Monitoring validation is mandatory.

TEST-SEC-05

Audit validation is mandatory.

TEST-SEC-06

Coverage must be measurable.

18.50 Architectural Constraints

AC-TEST-SEC-01

Untested security controls are prohibited.

AC-TEST-SEC-02

Access control validation cannot be skipped.

AC-TEST-SEC-03

Security defects require remediation.

AC-TEST-SEC-04

Coverage gaps require remediation.

AC-TEST-SEC-05

Verification evidence cannot be deleted.

AC-TEST-SEC-06

Governance controls cannot be bypassed.

18.51 Section Summary

This section defines the security testing strategy, security control validation requirements, monitoring verification approach, governance protections, audit requirements, resilience standards, and security compliance controls for TODP.

The security testing layer ensures that platform assets remain protected, auditable, governance-compliant, resilient, and resistant to unauthorized access, misuse, disclosure, corruption, and operational compromise.

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☐ Section 16 — Governance Module Test Specifications

☐ Section 17 — Audit & Replay Testing Specifications

☐ Section 18 — Security Testing Specifications

☐ Section 19 — Performance & Load Testing Specifications

TEST SPEC V1 — Section 19

Performance & Load Testing Specifications

Document:

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19

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19.1 Purpose

This section defines the performance and load testing specifications for TODP.

Performance testing validates the responsiveness, scalability, throughput, stability, and operational capacity of the platform under expected and extreme workloads.

The objective is to verify:

- Response time objectives
- Throughput objectives
- Scalability characteristics
- Resource utilization
- System stability
- Production readiness

19.2 Testing Objectives

PERF-TEST-01

Verify response times.

PERF-TEST-02

Verify throughput capacity.

PERF-TEST-03

Verify scalability.

PERF-TEST-04

Verify system stability.

PERF-TEST-05

Verify operational readiness.

PERF-TEST-06

Verify performance reproducibility.

19.3 Scope

Components Covered:

Frontend

Backend APIs

PostgreSQL

Redis

Retrieval Engine

Ranking Engine

Recommendation Engine

Governance Engine

Audit Engine

Infrastructure Components

19.4 Performance Validation Model

Workload

↓

Execution



Measurement



Analysis



Optimization



Verification

All performance objectives must be validated.

19.5 Performance Test Categories

Load Testing

Stress Testing

Spike Testing

Endurance Testing

Scalability Testing

Capacity Testing

19.6 Baseline Performance Tests

Verify:

Normal Workloads

Expected User Volumes

Expected Data Volumes

Expected Operational Patterns

19.7 Load Testing

Validate:

Expected Concurrent Users

Expected Request Volumes

Expected Data Volumes

Expected Throughput

19.8 Stress Testing

Verify:

Beyond-Normal Workloads

Resource Saturation

Graceful Degradation

Recovery Behavior

Failure Boundaries

19.9 Spike Testing

Validate:

Sudden Traffic Increases

Sudden Request Surges

Recovery Behavior

Elastic Scaling Behavior

19.10 Endurance Testing

Verify:

Long Duration Workloads

Memory Stability

Resource Stability

Operational Stability

Performance Consistency

19.11 Scalability Testing

Validate:

Horizontal Scaling

Vertical Scaling

Database Scaling

Cache Scaling

Service Scaling

19.12 Capacity Testing

Verify:

Maximum User Capacity

Maximum Throughput

Maximum Storage Capacity

Maximum Search Capacity

Maximum Recommendation Capacity

19.13 API Performance Tests

Validate:

Authentication APIs

Opportunity APIs

Evidence APIs

Retrieval APIs

Ranking APIs

Recommendation APIs

Governance APIs

Audit APIs

19.14 Authentication Performance Tests

Verify:

Login Latency

Token Validation Latency

Refresh Token Latency

Session Validation Latency

19.15 Retrieval Performance Tests

Validate:

Keyword Search Latency

Semantic Search Latency

Hybrid Search Latency

Replay Latency

History Retrieval Latency

19.16 Ranking Performance Tests

Verify:

Formula Execution Latency

Score Calculation Latency

Ranking Generation Latency

Replay Latency

19.17 Recommendation Performance Tests

Validate:

Recommendation Generation Latency

Explanation Generation Latency

Traceability Generation Latency

Replay Latency

19.18 Governance Performance Tests

Verify:

Policy Evaluation Latency

Approval Workflow Latency

Compliance Validation Latency

Decision Retrieval Latency

19.19 Audit Performance Tests

Validate:

Audit Write Latency

Audit Retrieval Latency

Snapshot Retrieval Latency

Replay Latency

19.20 Database Performance Tests

Verify:

Insert Performance

Update Performance

Delete Performance

Query Performance

Transaction Performance

19.21 Cache Performance Tests

Validate:

Cache Read Latency

Cache Write Latency

Cache Hit Ratio

Cache Miss Handling

Cache Scalability

19.22 Search Performance Tests

Verify:

Search Throughput

Search Accuracy Under Load

Result Consistency

Search Stability

19.23 Concurrent User Tests

Validate:

10 Users

50 Users

100 Users

500 Users

1000+ Users

19.24 Concurrent Request Tests

Verify:

Read Workloads

Write Workloads

Mixed Workloads

Search Workloads

Governance Workloads

19.25 Throughput Tests

Measure:

Requests Per Second

Transactions Per Second

Searches Per Second

Recommendations Per Second

Audit Writes Per Second

19.26 Resource Utilization Tests

Monitor:

CPU Usage

Memory Usage

Disk Usage

Network Usage

Database Usage

19.27 Performance Stability Tests

Verify:

Latency Stability

Throughput Stability

Resource Stability

Error Rate Stability

19.28 Bottleneck Identification Tests

Analyze:

Database Bottlenecks

API Bottlenecks

Search Bottlenecks

Infrastructure Bottlenecks

Network Bottlenecks

19.29 Recovery Tests

Verify:

Recovery After Load

Recovery After Stress

Recovery After Failure

Operational Restoration

19.30 SLA Validation Tests

Validate:

Latency Objectives

Availability Objectives

Throughput Objectives

Operational Objectives

19.31 Benchmark Tests

Establish:

Baseline Metrics

Target Metrics

Maximum Capacity Metrics

Production Readiness Metrics

19.32 Performance Workflow Tests

Validate End-To-End Flow:

User Request



API Processing



Retrieval



Ranking



Recommendation



Governance Validation



Audit Recording



Response Delivery

19.33 Positive Performance Tests

Verify:

Expected Load Handling

Expected Throughput

Expected Latency

Expected Stability

19.34 Negative Performance Tests

Verify:

Extreme Workloads

Resource Exhaustion

Unexpected Load Patterns

System Degradation

19.35 Boundary Performance Tests

Validate:

Maximum Users

Maximum Requests

Maximum Data Volumes

Maximum Search Volumes

Maximum Audit Volumes

19.36 Resilience Under Load Tests

Verify:

Graceful Degradation

Error Containment

Resource Recovery

Operational Continuity

19.37 Test Data Requirements

Use:

Synthetic User Data

Synthetic Opportunities

Synthetic Evidence

Synthetic Search Histories

Large Volume Datasets

19.38 Environment Requirements

Execute In:

Performance Test Environment

Staging Environment

Production-Like Environment

19.39 Coverage Targets

Performance Coverage:

100%

Critical Workflow Coverage:

100%

Scalability Coverage:

100%

Load Coverage:

100%

19.40 Defect Detection Goals

Detect:

Latency Defects

Scalability Defects

Throughput Defects

Resource Defects

Stability Defects

Capacity Defects

19.41 Pass Criteria

Performance Test Passes When:

Latency Targets Met

Throughput Targets Met

Stability Maintained

Resources Controlled

No Critical Failures

19.42 Failure Criteria

Performance Test Fails When:

Latency Exceeds Targets

Throughput Below Targets

Resource Exhaustion Occurs

Operational Stability Lost

Critical Errors Observed

19.43 Traceability Requirements

Every Performance Test Maps To:

Requirement



Performance Objective



Execution Result



Verification Record

19.44 Auditability Requirements

Maintain:

Performance Reports

Benchmark Reports

Coverage Reports

Defect Reports

Capacity Reports

Trend Reports

19.45 Reproducibility Requirements

Support:

Load Replay

Benchmark Replay

Execution Replay

Result Replay

Performance Replay

19.46 Governance Rules

TEST-PERF-01

All critical workflows require performance testing.

TEST-PERF-02

Capacity validation is mandatory.

TEST-PERF-03

Scalability validation is mandatory.

TEST-PERF-04

SLA validation is mandatory.

TEST-PERF-05

Performance evidence must be retained.

TEST-PERF-06

Coverage must be measurable.

19.47 Architectural Constraints

AC-TEST-PERF-01

Untested critical workflows are prohibited.

AC-TEST-PERF-02

Performance bottlenecks require remediation.

AC-TEST-PERF-03

Capacity validation cannot be skipped.

AC-TEST-PERF-04

Coverage gaps require remediation.

AC-TEST-PERF-05

Verification evidence cannot be deleted.

AC-TEST-PERF-06

Governance controls cannot be bypassed.

19.48 Section Summary

This section defines the performance testing strategy, load validation requirements, scalability verification approach, benchmarking controls, capacity validation standards, audit requirements, and production readiness criteria for TODP.

The performance testing layer ensures that the platform remains responsive, scalable, stable, efficient, auditable, governance-compliant, and capable of supporting expected and future operational workloads.

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TEST SPEC V1 — Section 20

Observability Testing Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

20

Version:

V1

Status:

APPROVED

20.1 Purpose

This section defines the testing specifications for platform observability.

Observability testing validates that TODP provides sufficient visibility into application behavior, infrastructure health, business workflows, governance activities, and operational incidents.

The objective is to verify:

- Logging correctness
- Metrics accuracy
- Trace completeness
- Alert effectiveness
- Operational visibility
- Governance visibility

20.2 Testing Objectives

OBS-TEST-01

Verify logging accuracy.

OBS-TEST-02

Verify metric accuracy.

OBS-TEST-03

Verify distributed tracing.

OBS-TEST-04

Verify alert generation.

OBS-TEST-05

Verify operational visibility.

OBS-TEST-06

Verify observability auditability.

20.3 Scope

Components Covered:

Application Logs

Audit Logs

Metrics Collection

Distributed Tracing

Dashboards

Alerting Systems

Monitoring Services

Operational Reporting

Governance Reporting

20.4 Observability Validation Model

System Event



Log Generation



Metric Generation



Trace Generation



Monitoring



Alerting



Investigation

All stages must be validated.

20.5 Logging Tests

Verify:

Log Creation

Log Persistence

Log Retrieval

Log Consistency

Log Completeness

20.6 Application Logging Tests

Validate:

API Logs

Service Logs

Repository Logs

Workflow Logs

Error Logs

20.7 Security Logging Tests

Verify:

Authentication Events

Authorization Events

Security Violations

Access Denials

Administrative Actions

20.8 Governance Logging Tests

Validate:

Policy Decisions

Approval Events

Compliance Events

Governance Actions

Replay Events

20.9 Audit Logging Tests

Verify:

Audit Record Logging

Snapshot Logging

Replay Logging

Traceability Logging

Correlation Logging

20.10 Structured Logging Tests

Validate:

Log Schema Compliance

Field Consistency

Metadata Completeness

Correlation IDs

Trace IDs

20.11 Metrics Collection Tests

Verify:

Metric Generation

Metric Aggregation

Metric Storage

Metric Retrieval

Metric Accuracy

20.12 Infrastructure Metrics Tests

Validate:

CPU Metrics

Memory Metrics

Disk Metrics

Network Metrics

Container Metrics

20.13 Application Metrics Tests

Verify:

Request Counts

Error Counts

Latency Metrics

Throughput Metrics

Workflow Metrics

20.14 Business Metrics Tests

Validate:

Search Counts

Ranking Counts

Recommendation Counts

Governance Decisions

Approval Statistics

20.15 Distributed Tracing Tests

Verify:

Trace Creation

Trace Propagation

Trace Retrieval

Trace Completeness

Trace Accuracy

20.16 Correlation Tests

Validate:

Request Correlation

Workflow Correlation

Audit Correlation

Replay Correlation

Cross-Service Correlation

20.17 Dashboard Tests

Verify:

Dashboard Availability

Dashboard Accuracy

Dashboard Refresh

Dashboard Completeness

Visualization Integrity

20.18 Alert Generation Tests

Validate:

Threshold Alerts

Error Alerts

Security Alerts

Governance Alerts

Infrastructure Alerts

20.19 Alert Accuracy Tests

Verify:

Correct Triggering

False Positive Control

False Negative Detection

Alert Context

Alert Traceability

20.20 Incident Visibility Tests

Validate:

Incident Detection

Incident Logging

Incident Alerting

Incident Correlation

Incident Reporting

20.21 Operational Visibility Tests

Verify:

Service Visibility

Database Visibility

Cache Visibility

Workflow Visibility

Infrastructure Visibility

20.22 Governance Visibility Tests

Validate:

Governance Dashboards

Compliance Dashboards

Approval Dashboards

Audit Dashboards

Replay Dashboards

20.23 Search Observability Tests

Verify:

Search Metrics

Search Logs

Search Traces

Search Alerts

Search Dashboards

20.24 Ranking Observability Tests

Validate:

Ranking Metrics

Ranking Logs

Ranking Traces

Ranking Alerts

Ranking Dashboards

20.25 Recommendation Observability Tests

Verify:

Recommendation Metrics

Recommendation Logs

Recommendation Traces

Recommendation Alerts

Recommendation Dashboards

20.26 Audit & Replay Observability Tests

Validate:

Replay Metrics

Replay Logs

Replay Traces

Replay Alerts

Replay Dashboards

20.27 Error Monitoring Tests

Verify:

Exception Tracking

Error Classification

Error Correlation

Error Reporting

Error Alerting

20.28 End-To-End Visibility Tests

Validate Flow:

User Request



API



Retrieval



Ranking



Recommendation



Governance



Audit



Response

Full visibility required.

20.29 Positive Observability Tests

Verify:

Expected Logs

Expected Metrics

Expected Traces

Expected Alerts

Expected Dashboards

20.30 Negative Observability Tests

Verify:

Missing Logs

Missing Metrics

Missing Traces

Alert Failures

Dashboard Failures

20.31 Boundary Observability Tests

Validate:

Large Log Volumes

Large Metric Volumes

Large Trace Volumes

High Alert Volumes

Long Histories

20.32 Concurrency Tests

Verify:

Concurrent Log Generation

Concurrent Metric Collection

Concurrent Trace Generation

Concurrent Alert Processing

Data Consistency

20.33 Performance Tests

Validate:

Log Ingestion Latency

Metric Collection Latency

Trace Processing Latency

Alert Delivery Latency

Dashboard Refresh Latency

20.34 Resilience Tests

Verify:

Monitoring Failures

Logging Failures

Tracing Failures

Alerting Failures

Graceful Recovery

20.35 Test Data Requirements

Use:

Synthetic Logs

Synthetic Metrics

Synthetic Traces

Synthetic Alerts

Synthetic Incidents

20.36 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Observability Validation Environment

Production-Like Environment

20.37 Coverage Targets

Logging Coverage:

100%

Metrics Coverage:

100%

Tracing Coverage:

100%

Alert Coverage:

100%

20.38 Defect Detection Goals

Detect:

Logging Defects

Metric Defects

Tracing Defects

Alerting Defects

Dashboard Defects

Visibility Defects

20.39 Pass Criteria

Observability Test Passes When:

Logs Accurate

Metrics Accurate

Traces Complete

Alerts Correct

Dashboards Accurate

20.40 Failure Criteria

Observability Test Fails When:

Logs Missing

Metrics Incorrect

Traces Incomplete

Alerts Incorrect

Dashboards Misleading

20.41 Traceability Requirements

Every Observability Test Maps To:

Requirement



Observability Component



Execution Result



Verification Record

20.42 Auditability Requirements

Maintain:

Execution Reports

Logging Reports

Metrics Reports

Tracing Reports

Coverage Reports

Defect Reports

20.43 Reproducibility Requirements

Support:

Log Replay

Metric Replay

Trace Replay

Execution Replay

Result Replay

20.44 Governance Rules

TEST-OBS-01

All critical workflows require observability validation.

TEST-OBS-02

Logging validation is mandatory.

TEST-OBS-03

Metrics validation is mandatory.

TEST-OBS-04

Tracing validation is mandatory.

TEST-OBS-05

Alert validation is mandatory.

TEST-OBS-06

Coverage must be measurable.

20.45 Architectural Constraints

AC-TEST-OBS-01

Untested observability controls are prohibited.

AC-TEST-OBS-02

Missing visibility requires remediation.

AC-TEST-OBS-03

Critical workflows must be observable.

AC-TEST-OBS-04

Coverage gaps require remediation.

AC-TEST-OBS-05

Verification evidence cannot be deleted.

AC-TEST-OBS-06

Governance controls cannot be bypassed.

20.46 Section Summary

This section defines the observability testing strategy, logging validation requirements, metrics verification approach, tracing controls, alerting standards, operational visibility requirements, and governance reporting controls for TODP.

The observability testing layer ensures that platform operations remain transparent, diagnosable, auditable, measurable, governance-compliant, and operationally manageable throughout the lifecycle of the system.

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☐ Section 19 — Performance & Load Testing Specifications

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☐ Section 21 — CI/CD Testing Specifications

TEST SPEC V1 — Section 21

CI/CD Testing Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

21

Version:

V1

Status:

APPROVED

21.1 Purpose

This section defines the testing specifications for Continuous Integration and Continuous Deployment (CI/CD) processes within TODP.

CI/CD testing validates that software changes move through build, validation, testing, security verification, deployment, and release workflows in a controlled, repeatable, auditable, and governance-compliant manner.

The objective is to verify:

- Build integrity
- Pipeline correctness
- Deployment reliability
- Release quality
- Governance compliance
- Operational reproducibility

21.2 Testing Objectives

CICD-TEST-01

Verify build processes.

CICD-TEST-02

Verify automated validation.

CICD-TEST-03

Verify deployment workflows.

CICD-TEST-04

Verify rollback capabilities.

CICD-TEST-05

Verify governance controls.

CICD-TEST-06

Verify pipeline reproducibility.

21.3 Scope

Components Covered:

Source Control

Build Pipelines

Test Pipelines

Security Pipelines

Artifact Repositories

Deployment Pipelines

Release Pipelines

Rollback Mechanisms

Infrastructure Provisioning

21.4 CI/CD Validation Model

Code Change

↓

Build

↓

Validation



Testing



Security Verification



Artifact Creation



Deployment



Release



Monitoring

All stages must be validated.

21.5 Source Control Validation Tests

Verify:

Branch Protection

Pull Request Validation

Merge Validation

Version Tracking

Commit Traceability

21.6 Build Pipeline Tests

Validate:

Backend Builds

Frontend Builds

Infrastructure Builds

Container Builds

Dependency Resolution

21.7 Build Integrity Tests

Verify:

Build Completeness

Build Consistency

Build Reproducibility

Artifact Integrity

Version Consistency

21.8 Automated Test Pipeline Tests

Validate:

Unit Test Execution

Integration Test Execution

API Test Execution

Database Test Execution

Regression Test Execution

21.9 Coverage Gate Tests

Verify:

Coverage Calculation

Coverage Thresholds

Pipeline Enforcement

Build Rejection Rules

Compliance Reporting

21.10 Static Analysis Tests

Validate:

Code Quality Analysis

Complexity Analysis

Lint Validation

Policy Validation

Quality Gate Enforcement

21.11 Security Pipeline Tests

Verify:

Dependency Scanning

Secret Detection

Container Scanning

Configuration Scanning

Security Gate Enforcement

21.12 Artifact Repository Tests

Validate:

Artifact Storage

Artifact Retrieval

Version Management

Integrity Verification

Retention Policies

21.13 Container Build Tests

Verify:

Container Creation

Container Validation

Container Security

Container Reproducibility

Version Tagging

21.14 Deployment Pipeline Tests

Validate:

Development Deployment

Testing Deployment

Staging Deployment

Production Deployment

Environment Validation

21.15 Deployment Verification Tests

Verify:

Service Availability

Database Connectivity

Cache Connectivity

Configuration Validity

Health Checks

21.16 Blue-Green Deployment Tests

Validate:

Environment Switching

Traffic Routing

Deployment Verification

Rollback Readiness

Operational Stability

21.17 Canary Deployment Tests

Verify:

Incremental Rollout

Traffic Segmentation

Monitoring Integration

Rollback Triggers

Risk Containment

21.18 Rollback Tests

Validate:

Application Rollback

Database Rollback

Configuration Rollback

Infrastructure Rollback

Version Restoration

21.19 Infrastructure Pipeline Tests

Verify:

Provisioning Execution

Infrastructure Validation

Configuration Consistency

Resource Integrity

Infrastructure Replay

21.20 Environment Promotion Tests

Validate:

Development → Testing

Testing → Staging

Staging → Production

Approval Gates

Promotion Traceability

21.21 Release Management Tests

Verify:

Release Packaging

Release Validation

Release Approval

Release Traceability

Release Auditability

21.22 Governance Gate Tests

Validate:

Approval Requirements

Security Approval

Quality Approval

Compliance Approval

Release Approval

21.23 Pipeline Failure Tests

Verify:

Build Failures

Test Failures

Security Failures

Deployment Failures

Recovery Procedures

21.24 Notification Tests

Validate:

Build Notifications

Deployment Notifications

Failure Notifications

Approval Notifications

Release Notifications

21.25 Monitoring Integration Tests

Verify:

Deployment Monitoring

Pipeline Monitoring

Release Monitoring

Alert Generation

Operational Visibility

21.26 Auditability Tests

Validate:

Pipeline Audits

Deployment Audits

Release Audits

Rollback Audits

Approval Audits

21.27 Traceability Tests

Verify:

Commit Traceability

Build Traceability

Artifact Traceability

Deployment Traceability

Release Traceability

21.28 Reproducibility Tests

Validate:

Build Replay

Pipeline Replay

Deployment Replay

Rollback Replay

Release Replay

21.29 End-To-End Pipeline Tests

Validate Flow:

Code Commit

↓

Build

↓

Testing

↓

Security Validation

↓

Artifact Creation

↓

Deployment

↓

Approval

↓

Release

↓

Monitoring

21.30 Positive Pipeline Tests

Verify:

Valid Commits

Successful Builds

Successful Tests

Successful Deployments

Expected Outcomes

21.31 Negative Pipeline Tests

Verify:

Broken Builds

Failed Tests

Failed Security Checks

Deployment Failures

Approval Violations

21.32 Boundary Pipeline Tests

Validate:

Large Builds

Large Deployments

Large Artifacts

Multiple Concurrent Releases

Complex Rollback Scenarios

21.33 Concurrency Tests

Verify:

Concurrent Builds

Concurrent Deployments

Concurrent Releases

Concurrent Rollbacks

Data Consistency

21.34 Performance Tests

Validate:

Build Duration

Pipeline Duration

Deployment Duration

Rollback Duration

Release Duration

21.35 Resilience Tests

Verify:

Infrastructure Failures

Pipeline Failures

Artifact Failures

Deployment Failures

Graceful Recovery

21.36 Test Data Requirements

Use:

Synthetic Commits

Synthetic Releases

Synthetic Deployments

Synthetic Rollbacks

Synthetic Artifacts

21.37 Environment Requirements

Execute In:

Testing Environment

Staging Environment

CI/CD Validation Environment

Production-Like Environment

21.38 Coverage Targets

Pipeline Coverage:

100%

Deployment Coverage:

100%

Rollback Coverage:

100%

Governance Coverage:

100%

21.39 Defect Detection Goals

Detect:

Build Defects

Pipeline Defects

Deployment Defects

Rollback Defects

Governance Defects

Traceability Defects

21.40 Pass Criteria

CI/CD Test Passes When:

Build Successful

Pipeline Successful

Deployment Successful

Rollback Successful

Governance Controls Enforced

21.41 Failure Criteria

CI/CD Test Fails When:

Build Failure Unhandled

Deployment Failure Unhandled

Rollback Failure Occurs

Governance Controls Bypassed

Traceability Lost

21.42 Traceability Requirements

Every CI/CD Test Maps To:

Requirement

↓

Pipeline Component

↓

Execution Result

↓

Verification Record

21.43 Auditability Requirements

Maintain:

Build Reports

Pipeline Reports

Deployment Reports

Release Reports

Coverage Reports

Defect Reports

21.44 Reproducibility Requirements

Support:

Build Replay

Deployment Replay

Release Replay

Rollback Replay

Execution Replay

21.45 Governance Rules

TEST-CICD-01

All deployment pipelines require testing.

TEST-CICD-02

Rollback validation is mandatory.

TEST-CICD-03

Security gate validation is mandatory.

TEST-CICD-04

Approval gate validation is mandatory.

TEST-CICD-05

Audit evidence must be retained.

TEST-CICD-06

Coverage must be measurable.

21.46 Architectural Constraints

AC-TEST-CICD-01

Untested deployment pipelines are prohibited.

AC-TEST-CICD-02

Rollback validation cannot be skipped.

AC-TEST-CICD-03

Security gates cannot be bypassed.

AC-TEST-CICD-04

Coverage gaps require remediation.

AC-TEST-CICD-05

Verification evidence cannot be deleted.

AC-TEST-CICD-06

Governance controls cannot be bypassed.

21.47 Section Summary

This section defines the testing strategy, build validation requirements, deployment verification approach, rollback controls, governance gates, audit requirements, and reproducibility standards for TODP CI/CD processes.

The CI/CD testing layer ensures that software delivery remains reliable, repeatable, auditable, traceable, governance-compliant, and operationally safe throughout the lifecycle of the platform.

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TEST SPEC V1 — Section 22

Disaster Recovery Testing Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

22

Version:

V1

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APPROVED

22.1 Purpose

This section defines the disaster recovery testing specifications for TODP.

Disaster Recovery (DR) testing validates the platform's ability to recover from infrastructure failures, service outages, database corruption, data loss events, security incidents, and operational disruptions while maintaining business continuity and governance compliance.

The objective is to verify:

- Recovery capabilities
- Backup integrity
- Failover readiness
- Data restoration
- Operational continuity
- Governance compliance

22.2 Testing Objectives

DR-TEST-01

Verify backup integrity.

DR-TEST-02

Verify restoration procedures.

DR-TEST-03

Verify failover capabilities.

DR-TEST-04

Verify recovery objectives.

DR-TEST-05

Verify operational continuity.

DR-TEST-06

Verify governance compliance.

22.3 Scope

Components Covered:

Application Services

PostgreSQL

Redis

Object Storage

Audit Systems

Governance Systems

Observability Stack

Infrastructure Components

CI/CD Systems

22.4 Disaster Recovery Validation Model

Failure Event



Detection



Containment



Recovery Activation



Restoration



Validation



Operational Recovery

All stages must be validated.

22.5 Backup Validation Tests

Verify:

Backup Creation

Backup Integrity

Backup Completeness

Backup Accessibility

Backup Retention

22.6 Database Backup Tests

Validate:

Full Backups

Incremental Backups

Transaction Log Backups

Backup Scheduling

Backup Verification

22.7 Application Backup Tests

Verify:

Configuration Backups

Artifact Backups

Deployment Backups

Infrastructure Backups

Version Backups

22.8 Audit Backup Tests

Validate:

Audit Record Backups

Snapshot Backups

Replay Backups

Governance Backups

Traceability Backups

22.9 Restore Validation Tests

Verify:

Database Restoration

Configuration Restoration

Application Restoration

Infrastructure Restoration

Audit Restoration

22.10 Database Restore Tests

Validate:

Schema Recovery

Data Recovery

Version Recovery

Constraint Recovery

Relationship Recovery

22.11 Point-In-Time Recovery Tests

Verify:

Historical Restoration

Recovery Accuracy

Recovery Consistency

Transaction Replay

Timeline Integrity

22.12 Failover Tests

Validate:

Primary Failure Detection

Secondary Activation

Traffic Switching

Operational Continuity

Recovery Validation

22.13 Service Recovery Tests

Verify:

API Recovery

Database Recovery

Cache Recovery

Monitoring Recovery

Governance Recovery

22.14 Infrastructure Failure Tests

Validate:

Server Failures

Container Failures

Network Failures

Storage Failures

Cloud Resource Failures

22.15 Database Failure Tests

Verify:

Database Crash Recovery

Corruption Recovery

Connection Recovery

Transaction Recovery

Replication Recovery

22.16 Cache Failure Tests

Validate:

Redis Failures

Cache Rebuild

Cache Recovery

Fallback Operations

Operational Continuity

22.17 Application Failure Tests

Verify:

Application Crash Recovery

Service Restart

Dependency Recovery

Workflow Recovery

State Consistency

22.18 Security Incident Recovery Tests

Validate:

Credential Compromise Response

Access Revocation

Secret Rotation

Audit Preservation

Recovery Validation

22.19 Governance Recovery Tests

Verify:

Policy Recovery

Approval Recovery

Compliance Recovery

Audit Recovery

Replay Recovery

22.20 Observability Recovery Tests

Validate:

Monitoring Recovery

Logging Recovery

Tracing Recovery

Alert Recovery

Dashboard Recovery

22.21 Recovery Objective Tests

Verify:

RTO Compliance

Recovery Process Accuracy

Recovery Consistency

Operational Readiness

Business Continuity

22.22 Recovery Point Tests

Validate:

RPO Compliance

Data Loss Limits

Backup Freshness

Recovery Accuracy

22.23 Operational Continuity Tests

Verify:

Critical Workflow Continuity

Governance Continuity

Audit Continuity

Monitoring Continuity

User Access Continuity

22.24 Data Integrity Recovery Tests

Validate:

Data Consistency

Relationship Integrity

Version Integrity

Audit Integrity

Governance Integrity

22.25 Disaster Simulation Tests

Verify:

Regional Outage Simulation

Infrastructure Outage Simulation

Database Failure Simulation

Application Failure Simulation

Security Incident Simulation

22.26 Recovery Workflow Tests

Validate End-To-End Flow:

Failure Detection

↓

Recovery Activation



Backup Retrieval



Restoration



Validation



Operational Recovery



Governance Validation

22.27 Positive Recovery Tests

Verify:

Successful Recovery

Successful Restoration

Successful Failover

Expected Outcomes

Operational Readiness

22.28 Negative Recovery Tests

Verify:

Corrupt Backups

Missing Backups

Recovery Failures

Invalid Recovery Requests

Governance Violations

22.29 Boundary Recovery Tests

Validate:

Large Databases

Large Audit Stores

Large Backup Sets

Long Recovery Timelines

Complex Failure Scenarios

22.30 Concurrency Tests

Verify:

Concurrent Recovery Requests

Concurrent Restores

Concurrent Failovers

Data Consistency

Transaction Integrity

22.31 Performance Tests

Validate:

Backup Duration

Restore Duration

Failover Duration

Recovery Duration

Validation Duration

22.32 Resilience Tests

Verify:

Repeated Failure Recovery

Recovery Under Load

Multi-Service Recovery

Cascading Failure Recovery

Graceful Recovery

22.33 Test Data Requirements

Use:

Synthetic Production-Like Data

Large Data Volumes

Historical Snapshots

Backup Archives

Recovery Scenarios

22.34 Environment Requirements

Execute In:

Disaster Recovery Environment

Staging Environment

Recovery Validation Environment

Production-Like Environment

22.35 Coverage Targets

Backup Coverage:

100%

Recovery Coverage:

100%

Failover Coverage:

100%

Governance Coverage:

100%

22.36 Defect Detection Goals

Detect:

Backup Defects

Restore Defects

Failover Defects

Recovery Defects

Governance Defects

Integrity Defects

22.37 Pass Criteria

Disaster Recovery Test Passes When:

Backups Valid

Recovery Successful

Failover Successful

Data Consistent

Governance Controls Preserved

22.38 Failure Criteria

Disaster Recovery Test Fails When:

Recovery Unsuccessful

Data Loss Exceeds Limits

Failover Fails

Governance Controls Lost

Audit Records Missing

22.39 Traceability Requirements

Every Disaster Recovery Test Maps To:

Requirement



Recovery Component



Execution Result



Verification Record

22.40 Auditability Requirements

Maintain:

Recovery Reports

Backup Reports

Restore Reports

Failover Reports

Coverage Reports

Defect Reports

22.41 Reproducibility Requirements

Support:

Recovery Replay

Restore Replay

Failover Replay

Execution Replay

Result Replay

22.42 Governance Rules

TEST-DR-01

All critical systems require disaster recovery testing.

TEST-DR-02

Backup validation is mandatory.

TEST-DR-03

Restore validation is mandatory.

TEST-DR-04

Failover validation is mandatory.

TEST-DR-05

Recovery evidence must be retained.

TEST-DR-06

Coverage must be measurable.

22.43 Architectural Constraints

AC-TEST-DR-01

Untested recovery procedures are prohibited.

AC-TEST-DR-02

Backup validation cannot be skipped.

AC-TEST-DR-03

Recovery objectives must be measurable.

AC-TEST-DR-04

Coverage gaps require remediation.

AC-TEST-DR-05

Verification evidence cannot be deleted.

AC-TEST-DR-06

Governance controls cannot be bypassed.

22.44 Section Summary

This section defines the disaster recovery testing strategy, backup validation requirements, restoration verification approach, failover controls, business continuity standards, governance protections, and operational recovery requirements for TODP.

The disaster recovery testing layer ensures that the platform remains recoverable, resilient, auditable, governance-compliant, and operationally reliable in the event of service disruptions, infrastructure failures, security incidents, and large-scale operational disasters.

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▢ Section 2 — Testing Strategy & Methodology

▢ Section 3 — Test Environment Architecture

▢ Section 4 — Unit Testing Specifications

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- 🔗 Section 23 — User Acceptance Testing (UAT)

TEST SPEC V1 — Section 23

User Acceptance Testing (UAT)

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

23

Version:

V1

Status:

APPROVED

23.1 Purpose

This section defines the User Acceptance Testing (UAT) specifications for TODP.

UAT validates that the platform satisfies business objectives, operational requirements, governance expectations, usability requirements, and stakeholder acceptance criteria before production deployment.

The objective is to verify:

- Business requirement fulfillment
- User workflow correctness
- Operational readiness
- Governance readiness

- Stakeholder acceptance
- Production readiness

23.2 UAT Objectives

UAT-01

Verify business requirements.

UAT-02

Verify end-user workflows.

UAT-03

Verify operational usability.

UAT-04

Verify governance compliance.

UAT-05

Verify stakeholder acceptance.

UAT-06

Verify deployment readiness.

23.3 Scope

Components Covered:

User Interface

Opportunity Management

Authority Management

Evidence Management

Retrieval Workflows

Ranking Workflows

Recommendation Workflows

Governance Workflows

Audit & Replay Workflows

Administrative Functions

23.4 UAT Validation Model

Business Requirement



User Workflow



User Execution



Result Verification



Acceptance Validation



Approval Decision

All business workflows must be validated.

23.5 Stakeholder Groups

Covered Stakeholders:

Business Users

Decision Makers

Governance Officers

Compliance Officers

System Administrators

Platform Owners

23.6 Business Requirement Validation

Verify:

Requirement Satisfaction

Expected Outcomes

Business Value

Workflow Alignment

Acceptance Criteria

23.7 User Workflow Validation

Validate:

Workflow Completion

Workflow Accuracy

Workflow Consistency

Workflow Traceability

Workflow Usability

23.8 Opportunity Workflow UAT

Verify:

Opportunity Creation

Opportunity Management

Opportunity Search

Opportunity Updates

Opportunity Lifecycle Operations

23.9 Authority Workflow UAT

Validate:

Authority Creation

Authority Validation

Authority Verification

Authority Search

Authority Lifecycle Operations

23.10 Evidence Workflow UAT

Verify:

Evidence Creation

Evidence Validation

Evidence Linking

Evidence Search

Evidence Lifecycle Operations

23.11 Retrieval Workflow UAT

Validate:

Keyword Search

Semantic Search

Hybrid Search

Search History

Search Replay

23.12 Ranking Workflow UAT

Verify:

Ranking Execution

Score Review

Ranking Review

Formula Transparency

Replay Operations

23.13 Recommendation Workflow UAT

Validate:

Recommendation Generation

Recommendation Review

Explanation Review

Traceability Review

Replay Operations

23.14 Governance Workflow UAT

Verify:

Policy Evaluation

Approval Processes

Compliance Validation

Governance Reporting

Governance Reviews

23.15 Audit Workflow UAT

Validate:

Audit Search

Audit Review

Replay Review

Historical Analysis

Traceability Review

23.16 Administrative Workflow UAT

Verify:

User Management

Role Management

Permission Management

Governance Administration

Platform Administration

23.17 Usability Validation

Validate:

Ease Of Use

Workflow Clarity

Navigation Clarity

Information Accessibility

Operational Efficiency

23.18 Accessibility Validation

Verify:

Keyboard Navigation

Screen Reader Compatibility

Readable Interfaces

Accessible Workflows

Accessibility Compliance

23.19 Data Accuracy Validation

Validate:

Displayed Data

Search Results

Rankings

Recommendations

Governance Decisions

23.20 Explainability Validation

Verify:

Recommendation Explanations

Ranking Explanations

Authority Explanations

Evidence Traceability

Governance Transparency

23.21 Traceability Validation

Validate:

Requirement Traceability

Decision Traceability

Evidence Traceability

Authority Traceability

Audit Traceability

23.22 Governance Validation

Verify:

Policy Enforcement

Approval Requirements

Compliance Requirements

Governance Reporting

Audit Requirements

23.23 Operational Readiness Validation

Validate:

Workflow Stability

System Availability

Performance Expectations

Support Readiness

Monitoring Readiness

23.24 Reporting Validation

Verify:

Business Reports

Governance Reports

Compliance Reports

Audit Reports

Operational Reports

23.25 User Feedback Collection

Capture:

Usability Feedback

Workflow Feedback

Performance Feedback

Governance Feedback

Improvement Requests

23.26 Defect Validation

Verify:

Reported Defects

Defect Resolution

Regression Verification

Acceptance Verification

Closure Validation

23.27 End-To-End UAT Scenarios

Validate Flow:

Opportunity Creation

↓

Evidence Linking

↓

Authority Verification

↓

Retrieval



Ranking



Recommendation



Governance Approval



Audit Review

23.28 Positive UAT Scenarios

Verify:

Expected Workflows

Expected Results

Expected Decisions

Expected Reports

Expected Governance Outcomes

23.29 Negative UAT Scenarios

Verify:

Invalid Inputs

Unauthorized Access

Workflow Failures

Governance Violations

Error Handling

23.30 Boundary UAT Scenarios

Validate:

Large Datasets

Complex Workflows

Multiple Governance Layers

High User Volumes

Long Operational Histories

23.31 Performance Validation

Verify:

Acceptable Response Times

Acceptable Workflow Completion Times

Acceptable Search Times

Acceptable Reporting Times

23.32 Environment Requirements

Execute In:

UAT Environment

Staging Environment

Production-Like Environment

Governance Validation Environment

23.33 Test Data Requirements

Use:

Representative Business Data

Synthetic Enterprise Data

Governance Scenarios

Compliance Scenarios

Historical Data

23.34 Coverage Targets

Business Workflow Coverage:

100%

Critical Workflow Coverage:

100%

Governance Coverage:

100%

Acceptance Coverage:

100%

23.35 Acceptance Criteria

UAT Passes When:

Business Requirements Met

User Workflows Successful

Governance Requirements Met

Stakeholder Approval Granted

Production Readiness Confirmed

23.36 Rejection Criteria

UAT Fails When:

Critical Workflow Failure Exists

Business Requirements Not Met

Governance Requirements Violated

Stakeholder Approval Denied

Production Readiness Not Achieved

23.37 Defect Detection Goals

Detect:

Workflow Defects

Usability Defects

Governance Defects

Reporting Defects

Acceptance Defects

Operational Defects

23.38 Traceability Requirements

Every UAT Scenario Maps To:

Business Requirement



Workflow



Acceptance Criteria



Execution Result

23.39 Auditability Requirements

Maintain:

UAT Reports

Acceptance Reports

Coverage Reports

Defect Reports

Approval Reports

Feedback Reports

23.40 Reproducibility Requirements

Support:

Workflow Replay

Scenario Replay

Execution Replay

Acceptance Replay

Result Replay

23.41 Governance Rules

TEST-UAT-01

All business workflows require UAT validation.

TEST-UAT-02

Stakeholder approval is mandatory.

TEST-UAT-03

Governance validation is mandatory.

TEST-UAT-04

Acceptance criteria must be measurable.

TEST-UAT-05

Acceptance evidence must be retained.

TEST-UAT-06

Coverage must be measurable.

23.42 Architectural Constraints

AC-TEST-UAT-01

Untested business workflows are prohibited.

AC-TEST-UAT-02

Stakeholder approval cannot be bypassed.

AC-TEST-UAT-03

Governance validation cannot be skipped.

AC-TEST-UAT-04

Coverage gaps require remediation.

AC-TEST-UAT-05

Verification evidence cannot be deleted.

AC-TEST-UAT-06

Production deployment without UAT approval is prohibited.

23.43 Section Summary

This section defines the User Acceptance Testing strategy, stakeholder validation requirements, workflow verification approach, governance acceptance controls, usability requirements, operational readiness criteria, and production acceptance standards for TODP.

The UAT layer ensures that TODP satisfies business objectives, governance requirements, operational expectations, stakeholder needs, and production readiness standards before platform release.

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❏ Section 23 — User Acceptance Testing (UAT)

❏ Section 24 — Test Data Management Specifications

TEST SPEC V1 — Section 24

Test Data Management Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

24

Version:

V1

Status:

APPROVED

24.1 Purpose

This section defines the Test Data Management (TDM) specifications for TODP.

Test Data Management ensures that testing activities use controlled, representative, traceable, secure, reproducible, and governance-compliant datasets across all testing environments.

The objective is to verify:

- Test data quality
- Test data consistency
- Test data traceability
- Test data security
- Test data reproducibility
- Governance compliance

24.2 Testing Objectives

TDM-01

Verify test data quality.

TDM-02

Verify test data integrity.

TDM-03

Verify test data traceability.

TDM-04

Verify test data security.

TDM-05

Verify test data reproducibility.

TDM-06

Verify governance compliance.

24.3 Scope

Components Covered:

Synthetic Data

Reference Data

Seed Data

Test Datasets

Scenario Datasets

Performance Datasets

Security Datasets

Governance Datasets

Audit Datasets

24.4 Test Data Lifecycle

Data Definition



Data Generation



Validation



Storage



Usage



Refresh



Retirement

All lifecycle stages must be controlled.

24.5 Test Data Categories

Supported Categories:

Functional Test Data

Integration Test Data

Performance Test Data

Security Test Data

Governance Test Data

Audit Test Data

UAT Test Data

Disaster Recovery Test Data

24.6 Synthetic Data Requirements

Verify:

Synthetic Generation

Data Realism

Data Consistency

Data Diversity

Data Repeatability

24.7 Seed Data Requirements

Validate:

Initial Dataset Creation

Reference Integrity

Relationship Integrity

Repeatability

Version Control

24.8 Reference Data Requirements

Verify:

Controlled Values

Lookup Values

Enumerations

Status Definitions

Governance Definitions

24.9 Test Data Validation

Validate:

Schema Compliance

Constraint Compliance

Relationship Integrity

Business Rule Compliance

Governance Compliance

24.10 Test Data Integrity Tests

Verify:

Data Completeness

Data Accuracy

Data Consistency

Relationship Accuracy

Version Consistency

24.11 Opportunity Test Data

Include:

Opportunity Scenarios

Lifecycle Scenarios

Search Scenarios

Version Scenarios

Governance Scenarios

24.12 Authority Test Data

Include:

Authority Scenarios

Verification Scenarios

Confidence Scenarios

Governance Scenarios

Audit Scenarios

24.13 Evidence Test Data

Include:

Evidence Scenarios

Lineage Scenarios

Relationship Scenarios

Version Scenarios

Audit Scenarios

24.14 Retrieval Test Data

Include:

Keyword Search Scenarios

Semantic Search Scenarios

Hybrid Search Scenarios

Replay Scenarios

Performance Scenarios

24.15 Ranking Test Data

Include:

Ranking Scenarios

Formula Scenarios

Weighting Scenarios

Replay Scenarios

Audit Scenarios

24.16 Recommendation Test Data

Include:

Recommendation Scenarios

Explanation Scenarios

Traceability Scenarios

Replay Scenarios

Governance Scenarios

24.17 Governance Test Data

Include:

Policy Scenarios

Approval Scenarios

Compliance Scenarios

Exception Scenarios

Audit Scenarios

24.18 Audit Test Data

Include:

Audit Scenarios

Replay Scenarios

Snapshot Scenarios

Correlation Scenarios

Historical Scenarios

24.19 Security Test Data

Include:

Authentication Scenarios

Authorization Scenarios

Attack Simulations

Invalid Inputs

Boundary Inputs

24.20 Performance Test Data

Include:

Large Datasets

High Volume Records

Large Search Histories

Large Audit Histories

Large Governance Histories

24.21 UAT Test Data

Include:

Representative Business Data

Operational Scenarios

Governance Scenarios

Acceptance Scenarios

Reporting Scenarios

24.22 Data Versioning Requirements

Verify:

Dataset Versioning

Dataset Retrieval

Dataset Replay

Version Consistency

Version Traceability

24.23 Data Refresh Requirements

Validate:

Refresh Scheduling

Refresh Consistency

Refresh Validation

Refresh Auditability

Refresh Traceability

24.24 Data Retention Requirements

Verify:

Retention Policies

Retention Periods

Archival Policies

Governance Requirements

Audit Requirements

24.25 Test Data Security

Validate:

Access Restrictions

Encryption Requirements

Storage Controls

Audit Logging

Data Protection

24.26 Sensitive Data Controls

Verify:

No Production Secrets

No Live Credentials

No Restricted Information

No Unauthorized Data

Approved Data Usage Only

24.27 Data Provisioning Tests

Validate:

Provisioning Accuracy

Provisioning Consistency

Provisioning Repeatability

Provisioning Automation

Provisioning Auditability

24.28 Data Cleanup Tests

Verify:

Cleanup Execution

Cleanup Validation

Data Isolation

Environment Reset

Audit Logging

24.29 Data Migration Tests

Validate:

Migration Accuracy

Migration Consistency

Relationship Integrity

Version Integrity

Audit Preservation

24.30 Dataset Reproducibility Tests

Verify:

Dataset Replay

Dataset Recreation

Dataset Version Replay

Scenario Replay

Result Consistency

24.31 End-To-End Data Workflow Tests

Validate Flow:

Dataset Definition

↓

Dataset Generation



Validation



Provisioning



Execution



Cleanup



Archival

24.32 Positive Data Tests

Verify:

Valid Datasets

Valid Relationships

Valid Scenarios

Expected Outcomes

Repeatable Results

24.33 Negative Data Tests

Verify:

Invalid Datasets

Missing Relationships

Corrupt Data

Invalid Formats

Policy Violations

24.34 Boundary Data Tests

Validate:

Large Datasets

Large Relationships

Large Histories

Maximum Scenario Volumes

Extreme Dataset Sizes

24.35 Concurrency Tests

Verify:

Concurrent Dataset Usage

Concurrent Provisioning

Concurrent Cleanup

Data Consistency

Isolation Integrity

24.36 Performance Tests

Validate:

Provisioning Speed

Refresh Speed

Migration Speed

Cleanup Speed

Retrieval Speed

24.37 Environment Requirements

Execute In:

Testing Environment

Staging Environment

Performance Environment

Governance Validation Environment

24.38 Coverage Targets

Dataset Coverage:

100%

Scenario Coverage:

100%

Governance Coverage:

100%

Replay Coverage:

100%

24.39 Defect Detection Goals

Detect:

Dataset Defects

Integrity Defects

Provisioning Defects

Security Defects

Governance Defects

Traceability Defects

24.40 Pass Criteria

Test Data Validation Passes When:

Datasets Valid

Relationships Correct

Security Controls Enforced

Governance Requirements Met

Reproducibility Confirmed

24.41 Failure Criteria

Test Data Validation Fails When:

Data Invalid

Relationships Broken

Security Controls Missing

Governance Violations Exist

Reproducibility Lost

24.42 Traceability Requirements

Every Dataset Maps To:

Requirement



Scenario



Execution Result



Verification Record

24.43 Auditability Requirements

Maintain:

Dataset Reports

Validation Reports

Coverage Reports

Provisioning Reports

Defect Reports

Audit Reports

24.44 Reproducibility Requirements

Support:

Dataset Replay

Provisioning Replay

Execution Replay

Scenario Replay

Result Replay

24.45 Governance Rules

TEST-TDM-01

All testing activities require approved datasets.

TEST-TDM-02

Dataset validation is mandatory.

TEST-TDM-03

Dataset security validation is mandatory.

TEST-TDM-04

Dataset traceability is mandatory.

TEST-TDM-05

Dataset evidence must be retained.

TEST-TDM-06

Coverage must be measurable.

24.46 Architectural Constraints

AC-TEST-TDM-01

Unvalidated datasets are prohibited.

AC-TEST-TDM-02

Production data usage is restricted.

AC-TEST-TDM-03

Dataset traceability cannot be skipped.

AC-TEST-TDM-04

Coverage gaps require remediation.

AC-TEST-TDM-05

Verification evidence cannot be deleted.

AC-TEST-TDM-06

Governance controls cannot be bypassed.

24.47 Section Summary

This section defines the test data management strategy, dataset validation requirements, provisioning controls, security protections, traceability standards, governance requirements, and reproducibility controls for TODP.

The Test Data Management layer ensures that testing activities are supported by reliable, secure, representative, auditable, governance-compliant, and reproducible datasets across all environments.

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- ❏ Section 24 — Test Data Management Specifications

- ❏ Section 25 — Test Reporting & Governance Specifications

TEST SPEC V1 — Section 25

Test Reporting & Governance Specifications

Document:

Test Specification (TEST SPEC V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

25

Version:

V1

Status:

APPROVED

25.1 Purpose

This section defines the Test Reporting and Test Governance specifications for TODP.

Test Reporting and Governance ensure that all testing activities are measurable, traceable, auditable, reproducible, and aligned with organizational quality standards, compliance obligations, and release approval requirements.

The objective is to verify:

- Test transparency
- Test traceability
- Test accountability
- Governance compliance
- Release readiness
- Auditability

25.2 Governance Objectives

TGOV-01

Verify testing completeness.

TGOV-02

Verify testing traceability.

TGOV-03

Verify testing accountability.

TGOV-04

Verify testing compliance.

TGOV-05

Verify release readiness.

TGOV-06

Verify testing auditability.

25.3 Scope

Components Covered:

Test Plans

Test Specifications

Test Cases

Test Executions

Coverage Reports

Defect Reports

Release Reports

Approval Records

Governance Records

Audit Records

25.4 Governance Validation Model

Requirement



Test Case



Execution



Evidence



Reporting



Approval



Release Decision

All stages must be traceable.

25.5 Test Reporting Requirements

Reports Must Include:

Execution Results

Coverage Results

Defect Results

Risk Assessments

Approval Status

25.6 Test Execution Reporting

Verify:

Execution Counts

Pass Counts

Fail Counts

Blocked Counts

Skipped Counts

25.7 Coverage Reporting

Validate:

Requirement Coverage

Code Coverage

API Coverage

Workflow Coverage

Governance Coverage

25.8 Defect Reporting

Verify:

Defect Identification

Defect Classification

Defect Prioritization

Defect Ownership

Defect Resolution Status

25.9 Defect Severity Levels

Supported Levels:

Critical

High

Medium

Low

Informational

25.10 Defect Metrics

Track:

Open Defects

Closed Defects

Reopened Defects

Defect Density

Defect Trends

25.11 Test Trend Reporting

Verify:

Execution Trends

Coverage Trends

Quality Trends

Defect Trends

Release Trends

25.12 Quality Metrics

Measure:

Pass Rate

Failure Rate

Coverage Percentage

Defect Escape Rate

Regression Stability

25.13 Risk Reporting

Validate:

Technical Risks

Security Risks

Governance Risks

Operational Risks

Release Risks

25.14 Governance Dashboard Requirements

Display:

Execution Status

Coverage Status

Defect Status

Risk Status

Approval Status

25.15 Release Readiness Reporting

Verify:

Quality Readiness

Security Readiness

Performance Readiness

Governance Readiness

Operational Readiness

25.16 Test Evidence Requirements

Maintain:

Execution Logs

Coverage Evidence

Defect Evidence

Approval Evidence

Audit Evidence

25.17 Traceability Reporting

Verify:

Requirement Traceability

Test Traceability

Execution Traceability

Defect Traceability

Approval Traceability

25.18 Audit Reporting

Validate:

Audit Records

Execution Audits

Approval Audits

Release Audits

Governance Audits

25.19 Approval Workflow Reporting

Verify:

Approval Requests

Approval Decisions

Approval History

Approval Integrity

Approval Traceability

25.20 Governance Review Requirements

Validate:

Quality Reviews

Security Reviews

Compliance Reviews

Release Reviews

Operational Reviews

25.21 Release Gate Validation

Verify:

Coverage Gates

Quality Gates

Security Gates

Governance Gates

Approval Gates

25.22 Compliance Reporting

Validate:

Policy Compliance

Security Compliance

Governance Compliance

Testing Compliance

Operational Compliance

25.23 Executive Reporting

Provide:

Quality Summaries

Risk Summaries

Defect Summaries

Release Summaries

Governance Summaries

25.24 Operational Reporting

Provide:

Execution Reports

Environment Reports

Performance Reports

Monitoring Reports

Operational Trends

25.25 Stakeholder Reporting

Support:

Engineering Teams

QA Teams

Governance Teams

Compliance Teams

Executive Stakeholders

25.26 Test Closure Reporting

Verify:

Execution Completion

Coverage Completion

Defect Resolution

Approval Completion

Release Readiness

25.27 Governance Workflow Tests

Validate Flow:

Requirement



Test Design



Execution



Evidence Collection



Reporting



Approval



Release Decision

25.28 Positive Governance Scenarios

Verify:

Successful Executions

Successful Approvals

Successful Releases

Expected Outcomes

Compliance Maintained

25.29 Negative Governance Scenarios

Verify:

Failed Executions

Missing Coverage

Critical Defects

Approval Failures

Compliance Violations

25.30 Boundary Governance Scenarios

Validate:

Large Test Suites

Large Defect Volumes

Large Coverage Reports

Multiple Releases

Complex Approval Chains

25.31 Concurrency Tests

Verify:

Concurrent Executions

Concurrent Reporting

Concurrent Approvals

Concurrent Releases

Data Consistency

25.32 Performance Tests

Validate:

Report Generation Speed

Dashboard Refresh Speed

Trend Calculation Speed

Approval Processing Speed

Data Retrieval Speed

25.33 Test Data Requirements

Use:

Execution Records

Coverage Records

Defect Records

Approval Records

Release Records

25.34 Environment Requirements

Execute In:

Testing Environment

Governance Validation Environment

Release Validation Environment

Production-Like Environment

25.35 Coverage Targets

Reporting Coverage:

100%

Governance Coverage:

100%

Approval Coverage:

100%

Release Coverage:

100%

25.36 Defect Detection Goals

Detect:

Reporting Defects

Governance Defects

Approval Defects

Traceability Defects

Compliance Defects

Audit Defects

25.37 Pass Criteria

Governance Validation Passes When:

Reports Accurate

Coverage Complete

Approvals Valid

Compliance Maintained

Release Readiness Confirmed

25.38 Failure Criteria

Governance Validation Fails When:

Reports Incorrect

Coverage Missing

Approvals Invalid

Compliance Violated

Release Readiness Not Achieved

25.39 Traceability Requirements

Every Governance Record Maps To:

Requirement



Test Artifact



Execution Result



Approval Record

25.40 Auditability Requirements

Maintain:

Execution Reports

Coverage Reports

Defect Reports

Approval Reports

Release Reports

Audit Reports

25.41 Reproducibility Requirements

Support:

Execution Replay

Report Replay

Approval Replay

Release Replay

Result Replay

25.42 Governance Rules

TEST-GOVREP-01

All testing activities require reporting.

TEST-GOVREP-02

Coverage validation is mandatory.

TEST-GOVREP-03

Approval validation is mandatory.

TEST-GOVREP-04

Release gate validation is mandatory.

TEST-GOVREP-05

Evidence must be retained.

TEST-GOVREP-06

Governance compliance must be measurable.

25.43 Architectural Constraints

AC-TEST-GOVREP-01

Unreported testing activities are prohibited.

AC-TEST-GOVREP-02

Coverage gaps require remediation.

AC-TEST-GOVREP-03

Approval bypass is prohibited.

AC-TEST-GOVREP-04

Release without governance approval is prohibited.

AC-TEST-GOVREP-05

Verification evidence cannot be deleted.

AC-TEST-GOVREP-06

Governance controls cannot be bypassed.

25.44 Section Summary

This section defines the test reporting strategy, governance validation requirements, traceability controls, approval workflows, release gate requirements, audit standards, and compliance reporting mechanisms for TODP.

The Test Reporting and Governance layer ensures that all testing activities remain transparent, auditable, traceable, measurable, governance-compliant, and aligned with organizational quality and release management standards.

TEST SPEC V1 Progress

📄 Section 1 — Document Overview & Testing Scope

📄 Section 2 — Testing Strategy & Methodology

- 🔗 Section 3 — Test Environment Architecture
- 🔗 Section 4 — Unit Testing Specifications
- 🔗 Section 5 — Integration Testing Specifications
- 🔗 Section 6 — API Testing Specifications
- 🔗 Section 7 — Database Testing Specifications
- 🔗 Section 8 — Authentication Testing Specifications
- 🔗 Section 9 — Authorization & RBAC Testing Specifications
- 🔗 Section 10 — Opportunity Module Test Specifications
- 🔗 Section 11 — Authority Module Test Specifications
- 🔗 Section 12 — Evidence Module Test Specifications
- 🔗 Section 13 — Retrieval Module Test Specifications
- 🔗 Section 14 — Ranking Module Test Specifications
- 🔗 Section 15 — Recommendation Module Test Specifications
- 🔗 Section 16 — Governance Module Test Specifications
- 🔗 Section 17 — Audit & Replay Testing Specifications
- 🔗 Section 18 — Security Testing Specifications
- 🔗 Section 19 — Performance & Load Testing Specifications
- 🔗 Section 20 — Observability Testing Specifications
- 🔗 Section 21 — CI/CD Testing Specifications
- 🔗 Section 22 — Disaster Recovery Testing Specifications
- 🔗 Section 23 — User Acceptance Testing (UAT)
- 🔗 Section 24 — Test Data Management Specifications
- 🔗 Section 25 — Test Reporting & Governance Specifications

- 🔗 TEST SPEC V1 COMPLETE (25/25 Sections)